

BEng (Hons) in Mechatronics and Autonomous Systems Engineering

Final Project

Design of a Mobile Signal Booster Drone

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possible.

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Abstract

In today's advanced telecommunications era, Mobile Network Operators (MNOs) offer cellular communication services with increasing effectiveness through multiple generations of technology. However, there are still areas with limited or no mobile network coverage, posing a significant challenge for communication in remote locations and tourist destinations. This lack of coverage can leave individuals This project aims to build a drone-mounted signal booster to address this problem. The project involves researching and developing signal amplifiers, positioning them with the assistance of a drone for optimal efficiency, and assessing the amplifiers' performance.The project's main objective is to build outside signal boosters and employ a drone to increase their range. However, there are drawbacks to take into account, such as manual drone operating requirements and the constrained amplification range of the gadgets.This project's implementation is anticipated to increase communication options for people in distant or emergency situations as well as mobile network coverage in those places. The initiative helps close the coverage gap in mobile networks and ensures connectivity even in difficult settings disconnected from the rest of the world, jeopardizing their safety in emergencies.

Chapter 1 - Introduction

1.1 Rational of the Topic

Mobile Network Operators (MNOs) provide their customers with cellular communications services such as access to the internet, data, SMS, and voice calls, Anywhere within their coverage areas. There were 5 generations of services introduced until 2023. The service capabilities had been improved with each generation to increase their effectiveness. In 1984, 1G was introduced. The highest bandwidth that could be used with this analog communication mode was 2.4 kbps. With a maximum bandwidth of 64 Kbps, 2G was introduced in 1999. 3G was introduced in 2001, making it possible to browse the web, send emails, and share photos. In 2010, 4G was released to the world, revolutionizing how communications are used today. 4G has a 1Gbps maximum bandwidth (1,048,576kb). 2019 saw the launch of 5G in the USA, which is 35 times faster than 4G.

It's unfortunate to acknowledge that there are areas without even Mobile Network coverage as the world advances in telecommunications such as Artificial Intelligence, the Internet of Things, high-definition mobile TV, video conferencing, etc. Despite the claims of mobile network providers that their network is fully covered across the nation, "No Service" signals are frequently seen as you move throughout the country. Numerous countries experience signal loss, yet the data indicates that there are still many areas where mobile network providers do not offer service. Tourism is a way Sri Lanka earns Millions of dollars because Sri Lanka is a country of exquisite beauty. Additionally, the majority of camping grounds, forests, and tourist destinations lack signal service. It is a serious drawback. Mobile is often the only form of communication with the outside world in most of those locations. The absence of Mobile Network Coverage in these areas is the same as they're disconnected from the rest of the world.

Nowadays, the majority of mobile phones cannot use 2G mobile network service. Consequently, smartphones display the message "No Service" when they enter a 2G only area. Only older smartphones have 2G coverage, and few people nowadays use these phones. Therefore, travelers are entirely lost in an emergency. Jungles, hiking trails, and camping grounds are among the locations that can pose a threat. Several risks can befall there, including snakes, elephants, animal attacks, forest fires, cold and windy nights, and many others. What if the best course of action in those circumstances is to call an ambulance or connect with someone who can assist but cannot connect due to signal loss.

The base station, which is used by mobile network providers, is a tall cell tower joined with antennas. These base stations' transmissions are what link our phones to the internet. If there was a means to improve the signal, the problems described above may be avoided. Mobile phone signals can be shown as 3G or 4G by using a signal amplifier to magnify the mobile network signal within a close range. This would allow for clearer and quicker connections as well as faster access through calls. However, in "No Service" locations, the likelihood of an amplifier providing good service is limited. So that the mobile device may connect, a method of amplifying and receiving signals from the tower is needed.

1.2 Problem Identification

There are many tourist-attracted places in Sri Lanka without network coverage. Animal attacks, cold and windy weather, forest fires, accidents, and other risks are present for hikers, campers, photographers, tourists, and other visitors. When they are cut off from the outside world, they are powerless in an emergency. However, if they had a reliable network connection, getting assistance may save their lives.

1.3 Aim and Objective

1.3.1 Aim

- To create two signal boosters and attach them to a drone and controller.

1.3.2 Objectives

- To research for the signals and signal amplifying
- To design signal amplifiers
- To control the positions of the Amplifiers to receive maximum results using a Drone
- To calculate the effectiveness of the amplifier

1.4 Scope and Limitations

1.4.1 Scope

- The Project idea is to create a signal boosters for outside use and expand the range of booster using a drone.

1.4.2 Limitations

- Manual control - User has to Manualy Operate the Drone
- Amplifies only in a limited area - Device is capable of only boosting inside a limited range

Chapter 2 - Literature Review

In this chapter, Mobile Networks, signal amplification, and designs are detailed.

2.1 Network Coverage in Sri Lanka

Sri Lanka's telecommunications landscape comprises four major network providers: Airtel, Dialog, Hutch, and Mobitel. Analyzing the data reveals that Airtel subscribers enjoy a commendable connectivity rate of 92.4% to a 3G signal or higher. However, Airtel's 4G network exhibits limited geographical coverage, scoring a modest 3.8 out of 10 on the 4G Coverage Experience scale. In stark contrast, Dialog emerges as the frontrunner with an impressive score of 8.9, indicating its dominant position in terms of expansive network coverage throughout the country. On Mobitel's network, Sri Lanka had the fastest average download speed with 9.6 Mbps, which was 1.1 Mbps (12.9%) faster than Dialog in second place. With a rating of 4.8 Mbps, dialog upload speed experience. Mobitel comes in second, lagging by 1.1 Mbps. As a result of its scores of 49.1% for Excellent Consistent Quality and 76.6% for Core Consistent Quality, Dialog takes up the Consistency award, beating out Airtel in second place.

Airtel is the only recipient of the availability award, as 92.4% of the time, customers on this network were connected to a 3G signal or greater. Statistically, Dialog, Hutch, and Mobitel are tied for second position with scores between 89.3 and 90.3%. Researchers also keep track of the situations that irritate mobile customers the most, such as when there is absolutely no signal to connect to. Users deal with dead zones most frequently inside. Since the majority of currently available data is gathered indoors (where users spend the majority of their time). In contrast to the user-centric and geographically-based technique utilized by researchers' reach measures, our availability measurements adopt a user-centric, time-based methodology.
The availability indicator displays the percentage of time that every Open signal user on a certain operator's network has a 3G, 4G, or 5G connection.

The coverage maps display the areas where users connected with 3G or greater mobile service provided measurements. Each map indicates the locations where customers of that mobile operator may access mobile service.

With an overall rating of 8.9 on a scale of 10, Dialog beats out second-placed Mobitel by one point (7.9) in terms of 4G coverage. According to Dialog's findings, roughly nine out of ten places visited by users of all network operators had 4G connectivity for our users on this network. Hutch took third place with 4.7 points. Mobile subscribers' experiences with 4G coverage on an operator's network are measured by 4G Coverage Experience. It compares the places visited by users of all network operators with those where consumers of a particular network operator obtained a 4G signal on a scale of 0 to 10. Every area that matters to regular users, including all of their places of residence, employment, and travel, is measured by 4G Coverage Experience. The total number of places visited by users, the percentage of those where 4G service is accessible, and the value of those locations to those users are all taken into account.

Image 1 A - Network Coverage

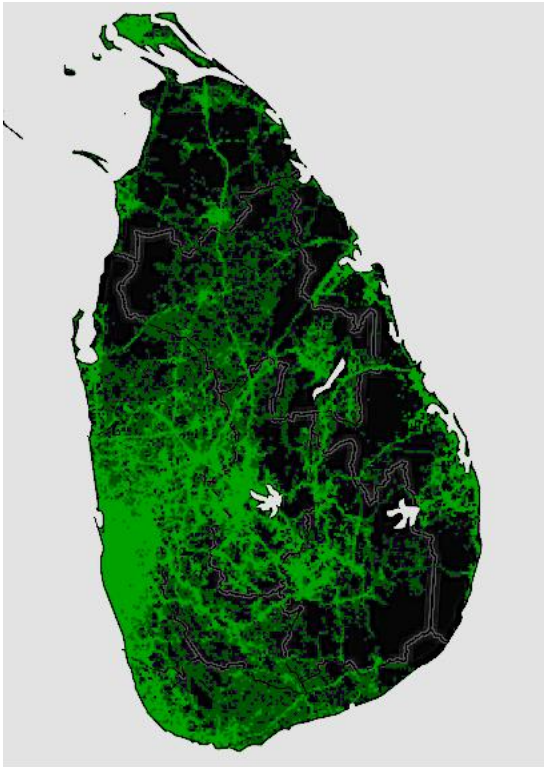


Image 2 B Tourist Attracted Places



Figure 1 Network Coverage and Tourist attracted places

The gathered data (2022-06) was used to create 4 map references for each mobile network's coverage and map A above displays all of the country's coverage regions for all 4 mobile networks (Airtel, Dialog, Hutch, and Mobitel). The pink areas are where all four networks have coverage, while the white parts are where they haven't. All of the white areas are assumed to have 2G or slower network signals. The pink sections contain very few places with 4G coverage. Map B is a comprehensive map of Sri Lanka that shows the country's tourism hotspots.

Operator	Technology	Download Speed (Mbps)	Upload Speed (Mbps)	Latency (ms)
Dialog	**4G**	**10.1**	**4.8**	**35**
Mobitel	**4G**	**9.6**	**4.5**	**36**
Hutch	**4G**	**8.1**	**3.2**	**40**
Airtel	**4G**	**7.2**	**2.8**	**45**

Table 1 comparison between service providers

The whole region of the national forests of Yala, Lunugamwehera, and Kumana is marked as "no service" when comparing the A and B maps. These National Forests are essential for wildlife-related activity and tourism. In an emergency, people who reside in these locations, researchers, travelers, photographers, and other people are at grave risk. The surrounding territory of Wilpattu National Park is not adequately covered. The regions of Mannar, Kilinochchi, Mullativu, and Vavuniya are also depicted in white to indicate the extent of coverage. All travelers, both local and foreign, visit Mannar Island, Mannar Fort, Adam's Bridge, the shrine of Our Lady Madhu from the Mannar district, Iranamadu Tank, Murukandy Pilleyar Kovil from Kilinochchi, Kaludiya Pokuna Forest, Mihintale Peaks, Wasagamuwa National Park, the Archeological Museum of Vavuniya, the Grand Jummah Mosque towers from Vavuniya.

2.2 Mobile Networks - 2G,3G,4G and 5G

A mobile network operator (MNO), also known as a carrier service provider is a company that offers telecommunications services to mobile subscribers who have subscribed to its network. Allowing them to call, text, and much more. First analog communication is known as 1G. These 1G phones had poor battery life, less security, dropped calls, etc. And years later 2G was created with digital telecommunication. 1G used Analog signals and 2G used digital signals. The main focus of the 2G was to reliable and secure communication channel. SMS and MMS data services were provided with 2G. The technical leap from 1G to 2G brought about most of the core services that we still use today, including SMS, internal roaming, video conferences, call holding, and subscription pricing, including real-time pricing and rates for long-distance calls. The top 2G speeds for General Packet Radio Service (GPRS) and Enhanced Data Rates for GSM Evolution, respectively, are 50 Kbps and 1 Mbps (EDGE). Before the important switch from 2G to 3G wireless networks, the less well-known 2.5G and 2.75G wireless network technologies functioned as a bridge.

To transmit signals across long distances, 3G makes use of a network of phone towers. This guarantees a continuous and speedy connection. The tower closest to the mobile device receives data. New frequency bands and greater data transfer are features of every generation. When 3G was first introduced, it was revolutionary because it allowed consumers to access the internet on their capable mobile devices, just as 2G did for text messaging. The third generation defined the norms for the great bulk of wireless technologies that we currently take for granted. The third generation of smartphones includes features including email, video downloads, photo sharing, and internet browsing. Third-generation mobile communication first became available commercially in 2001, with the stated aims of enabling more voice and data capacity, allowing a larger range of applications, and boosting data transmission at a lower cost. Through 3G, streaming and support for multimedia services are becoming more and more common. The success of 3G is made possible by universal access and mobility across different device types (telephones, PDAs, etc.). 3G increased the efficiency of the frequency spectrum so that more simultaneous calls could be made in the same frequency range by improving the way audio is compressed during a call. The

International Telecommunications Union IMT-2000 standard, which demands fixed speeds of 2 Mbps and mobile rates of 384 kbps, must be followed by a "true" 3G network. The maximum theoretical speed for HSPA+ is 21.6 Mbps. 3G evolved into 3.5G and 3.75G like how 2G did when more features were added to build 4G, resulting in 3G and 3.75G. A 4G phone can communicate over a 3G or even a 2G network, but a 3G phone cannot through a 4G network. Newer generations of phones are almost usually built to be backward compatible.

Both TDMA and CDMA are used in 3G technology. Utilizing value-added services like mobile television, GPS, and video conferencing, 3G (Third Generation Technology) technologies are used. Fast data transfer rates are the primary characteristic of 3G Technology However because ITU 200 is still deciding how to adjust the data rates, this functionality is not currently operating as intended. 2 Mbps is anticipated for stationary users, and 348 kbps for users who are moving or traveling. To employ broadband technology, ITU sells a variety of frequency rates. Users trust network authentication because they know their network is a dependable source for data transmission. The fact that 3G technology can handle all 5 of the main radio technologies makes it far more adaptable. These radio systems work with CDMA, TDMA, and FDMA. IMT-DS (direct spread) and IMT-MC are supported by CDMA (multi-carrier). IMT-TC and IMT-SC are covered by TDMA (single carrier). IMT-FC, often known as the frequency code, is the sole radio interface used by FDMA. Due to industrial cooperation, third-generation technology is rather inexpensive. This agreement was implemented quickly to boost user acceptance. The 3G (Third Generation Technology) network may operate alongside 2G networks. The third generation of technologies, or 3G, has the goal of being extensible on demand. With less investment, 3G aimed to provide greater coverage and development.

The development of 4G technology, which represents a major improvement over 3G, has been made possible chiefly by recent technological developments. Its implementation seeks to improve security, lower phone, and data service costs, and simplify multimedia and IP-based internet access. The major goal is to provide consumers with rapid, trustworthy, and high-capacity services. A few more potential uses of 4G include enhanced mobile web browsing, IP telephony, gaming services, high-definition mobile TV, video conferencing, 3D television, and cloud computing.

These developments provide a wealth of fascinating opportunities for the present and the future.

With high-speed upload and download packets, 4G provides customers with access to broadband-style speeds on their mobile devices, tablet, or laptop. With towers dispersed over the whole country, it operates effectively like a radio system. Your 4G-enabled device communicates with the base station (mast), which then sends data from the internet to the smartphone and vice versa. MIMO (Multiple Input Multiple Output) and OFDM are the main technologies that have made this possible (Orthogonal Frequency Division Multiplexing).LTE and WiMAX, the two main 4G technologies, are no longer being used. Long Term Evolution (LTE), a series of improvements to the present UMTS technology, will be used on Telstra's existing 1800 MHz frequency range. A 4G network enables maximum rates of up to 100 Mbps while a device is in motion. However, the internet connection functions admirably while the consumer remains motionless or moving around, with a bandwidth that may exceed 1 Gbps. Additionally, 4G networks are built to handle larger data volumes, which leads to less congestion and slicker data transfers. Reduced latency is the noticeable benefit, with 4G networks reporting a significant reduction from about 300 ms to about 100 ms. In plainer terms, 4G technology gave a substantial speed gain over 3G when it initially appeared. Despite being very close to meeting the standards' criteria, 4G LTE is not the same as 4G.

A 4G phone may connect to a 3G or even 2G network thanks to the backward compatibility of newer phone generations. Most operators seem to agree that OFDM is one of the crucial indicators that a service can legitimately be marketed as 4G. OFDM, which splits a signal into several narrowband channels at differing frequencies, is an illustration of digital modulation. Due to the circuit-switched nature of voice calls in GSM, UMTS, and CDMA2000, carriers will need to restructure their voice call network to enable LTE. Service providers will need to make a large investment in new infrastructure for this.

The Internet Protocol (IP), which uses a standard communications protocol to send and receive data in packets, is what 4G networks rely on, much as 3G networks. However, unlike 3G, 4G uses IP even for voice data. It is a standard for all IPs. The

data can travel across all types of networks using these standardized packets without being jumbled or tampered with. The phone must first connect to a base station to send and receive packets. A base station simply refers to those towering cell towers with various antenna attachments; it transfers data to and from the Internet and mobile devices. There are several ways (known as air interfaces) to connect the base station and phone. Each of the interfaces uses radio waves in a specific frequency to transmit data in various ways. All air interfaces have a crucial component: they manage network traffic, instructing devices on when to send and receive data. However, the methods they use to complete this duty vary.

2.3 Air Interface

The communication channel between the two stations in mobile or wireless communication is called the air interface, or access mode. The definition of the radio transmission between the transmitter and the receiver is referred to as the "air interface." The air interface specifies the modulation technique, multiple access strategy, pairing of frequencies (duplex distance), center frequencies, the bandwidth of radio channels, and other radio technology-related terms.

2.3.1 2G - GSM

The type of modulation utilized is simple to produce, enables the handset power amplifiers to function well, and allows for the transfer of enough data. Any radio-based system's fundamental component is the radio signal's format. Gaussian Minimum Shift Keying,(GMSK) a kind of phase sift keying, is used to modify the carrier. The GSM signal utilizing GMSK has a nominal bandwidth of 200 kHz. Due to the use of GMSK modulation, neighboring channels can be utilized from the same base station because the undesirable or spurious emissions beyond the nominal bandwidth are at a suitably low level. Usually, a number of carriers are allotted to each base station so that it can reach the necessary capacity. Under the fundamental system, the carrier's data is divided into eight-time slots to serve up to eight separate users. despite the basic carrier being able to sustain a data throughput of about 270 kbps. This is because the portion of this supports the management overhead.

A TDMA technique is then used to split the carriers into time segments. As a result, distinct time slots can be assigned to the various users of the same radio frequency channel. Each GSM burst, as well as each GSM slot, lasts for 15/26 mS, or 0.577 mS. The eight of these burst periods that make up a TDMA frame are clustered together. This serves as the fundamental building block for the formation of logical channels, lasting around 4.615 milliseconds (i.e. 120/26 ms). Each TDMA frame allows one burst period per physical channel. The lowest available time period for any mobile device is the GSM slot. Because a range of various sorts of data must be conveyed, it has a set format. 148 bits of information are typically sent using the slots. Voice data, control data, and synchronization data can all be carried via this data. The timing of the slots in the uplink and downlink are not synchronous, as can be observed from the GSM slot structure, and there is a time difference between the transmit and receive. This intentional offset in the GSM slot scheduling prevents a mobile device that is given the same slot in both directions from simultaneously transmitting and receiving. As a result, less costly filters are required to separate the transmitter and receiver.

2.3.2 3G - WCDMA - UMTS

The air interface standard for third-generation (3G) mobile networks that are most extensively used is WCDMA. The second-generation (2G) network (GSM) has been replaced by third-generation (3G) networks. The primary improvements between the two networks are:

• Higher user peak data rates (several Mbps).
• Dynamic user data rate adaption (also known as "bandwidth on demand").
• Support for differentiation in Quality of Service (QoS), such as in terms of error rates and latency.
• Multiplexing, which is the use of a single connection to deliver services with various quality and data rate needs.
• Enhanced spectrum effectiveness (higher system throughput per unit of allocated frequency spectrum).

Wideband CDMA, often known as WCDMA, is the radio transmission technology used by UMTS. It uses a channel bandwidth of 5 MHz. It can carry over 100 phone conversations at once with this bandwidth, or it can send data at rates of up to 2 Mbps in its original format. However, the data transfer rates have since grown to 14.4 Mbps thanks to additional improvements included in newer editions of the standard. To allow numerous phones or used appliances to have access to the base station, 3G UMTS uses the CDMA, or code division multiple access, protocol. The core 3G UMTS cellular architecture allowed for data speeds of up to 2048 kbps. However, as data usage grew quickly, these numbers became insufficient, necessitating further increases in data rate. To boost the downlink speed, a system known as HSDPA, high-speed packet download access, was initially deployed. HSUPA, or high-speed packet uplink access, was introduced after that.

A key component of the entire 3G UMTS network design was the USER Equipment or UE. It served as the user's ultimate interface. Although it had access to considerably faster data connectivity, it was still primarily a handset that could be much more adaptable and hold a wide range of apps.

The 3G UMTS core network architecture was a conversion of the GSM design with new components added to support the extra capabilities required by UMTS. The UMTS core network was divided into two parts because of the many methods that data may be transmitted:

Circuit-switched elements: These components, which largely draw their inspiration from GSM network entities, transmit data using a persistent channel for the length of the conversation.

Packet-switched elements: The purpose of these network components, known as packet-switched elements, is to transport packet data. Because capacity could be shared and data was transmitted in packets that were routed according to their destinations, this allowed for substantially higher network utilization.

2.3.3 4G , LTE -

Higher layers can access data transport services provided by the LTE air interface physical layer. Through the use of a transport channel over the MAC sub-layer, access to these services is possible. To offer the data transmission service, the physical layer must carry out the following tasks:

- Detecting errors on the transport channel and signaling them to higher levels
- Decoding and encoding of the transport channel using FEC
- ARQ soft-combining hybrid
- The coded transport channel mapping onto physical channels.
- Weighting of physical conduits for power
- Physical channel modulation and demodulation
- Time and frequency synchronization
- observations of radio properties and signals pointing to higher layers
- processing using MIMO (multiple input, multiple output) antennas
- Spread Diversity (TX diversity)
- RF processing
- Beamforming

LTE uses a variety of logical, transport, and physical channel types that may be identified by the information they carry and how that information is handled.

Generation	Range
1G	100 kbps to 1.2 kbps
2G	64 kbps to 144 kbps
3G	2 Mbps to 7.2 Mbps
4G	100 Mbps to 1 Gbps
5G	1 Gbps to 10 Gbps

Table 2 Generations and Range

A cell can be set up with a variety of bandwidths.

For instance, in a 5MHz band, BS uses 25 RBs to broadcast (equivalent to 12 x 25 = 300 sub-carriers), resulting in a transmission bandwidth of 4.5MHz (= 25 x 0.18MHz).

This configuration allows for guard bands to be placed at the upper and lower frequency band's boundaries (5MHz - 4.5MHz = 0.5MHz = 2 x 0.25MHz), which reduces interference with the band after it.

The network operator can change the breadth of the two guard bands as necessary by altering the center frequency by increments of 100 kHz.

All of these bandwidth options make it simple for network operators to deploy LTE in a variety of spectrum management regimes, for example, 1.4 MHz is comparable to the cdma2000 and TD-SCDMA bandwidths, 5 MHz is the same bandwidth as WCDMA, and 20 MsHz enables an LTE BS to operate at its maximum data rate.

2.3.4 4G signal Strength

>= -70 dBm	**Excellent**
-70 dBm to -85 dBm	**Good**
-86 dBm to -100 dBm	**Fair**
< -100 dBm	**Poor**
-110 dBm	**No signal**

Table 3 RSRP and RSRQ

2.4 Signal Boosting

Few techniques have been developed by organizations for signal boosters with the method of current outside signal is drawn in by an external antenna In the signal booster, which then amplifies and rebroadcasts it into the appropriate locations with a indoor antenna. The boosted signal is instantly picked up by cellular devices. deliverable texts, missed calls, and inaccurate data will be minimized or removed.
Additionally, there are various low-cost techniques for creating a small antenna that is wired together. There are signal boosters, signal repeaters, and antenna amplifiers available, but given the situations, a circuit-based, wireless, small-sized device with signal boosting capabilities is what required.

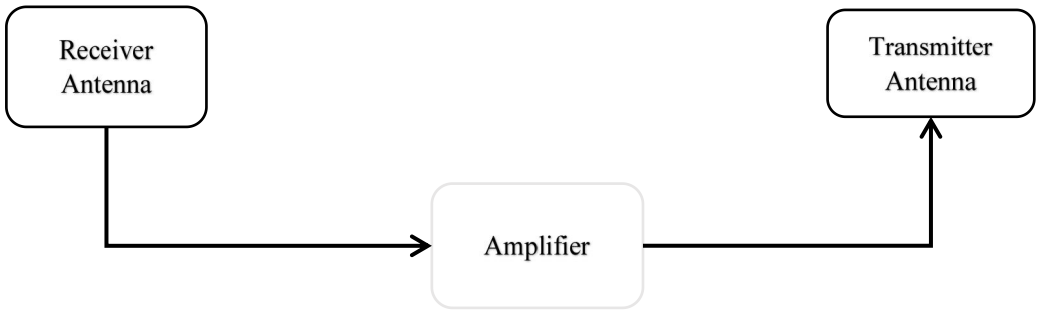


Figure 2 Simplified Diagram

According to the data gathered, there are few techniques to use a crystal oscillator to create signal boosters that function effectively. Although accuracy and range are still unknown, this solution satisfies a possible need for the current issue. By receiving a voltage signal from the quartz resonator, amplify it, and sending it back to the resonator, the crystal oscillator circuit maintains oscillation. The cut and size of the crystal define the resonance frequency, which is the rate of quartz expansion and contraction. An oscillation can be sustained when the energy of the produced output frequencies equals the circuit losses. Using this method to feed GSM, CDMA or LTE signals to the crystal oscillator and amplify it, and sending it back with a Antenna could create a ideal signal booster. On the principle of the inverse piezoelectric effect, crystal oscillators work by applying an alternating voltage to the crystal's surfaces, which causes the crystal to vibrate at its natural frequency. These vibrations are what

ultimately transform into oscillations.Even some other materials like Rochelle salt and tourmaline show the piezoelectric effect, these oscillators are often manufactured using quartz crystal since it is less costly, more readily accessible naturally, and more robust than other materials. The crystal oscillators' normal operating range is 40 KHz to 100 MHz, with the low-frequency oscillators being created with OpAmps and the high-frequency oscillators being designed with transistors (BJTs or FETs).

2.5 Drone

The increasing interest in UAVs has led to advancements in manufacturing technology, expanding their applications in areas such as search and rescue, parcel distribution, and telecommunications. UAVs are emerging as a promising technology for the future. It is important to secure database servers and incorporate additional devices when using UAVs over large areas. In rescue scenarios, a high-performance UAV network with repeaters and surveillance cameras is necessary for efficient data processing and successful mission completion.

The integration of UAVs with the 5G network holds great potential for improved monitoring and connectivity in various experiments. However, challenges arise when integrating UAVs with 5G ground base stations (5G-BSs), as equipment interference increases. Researchers are exploring new techniques to ensure seamless integration between aerial and land equipment within the same system. The need for efficient data transmission during crises and epidemics, such as the recent coronavirus outbreak, has highlighted the potential of UAVs as an alternative transport system integrated with radio signal boosters in epidemic areas.

Emergency communication networks play a vital role in rescue operations and exceptional circumstances when the integrated communication system is compromised. However, existing approaches often lack durability and have limitations in terms of space and environment. To overcome these weaknesses, this study proposes the use of unmanned aerial vehicles (UAVs), 5G-BSs, UAV signal

boosters, and database servers to establish an effective wireless coverage area with stability and mobility advantages.

The specific focus of this study is on utilizing UAVs equipped with mobile signal booster. The weak wireless signals in remote emergency areas are strengthened by a wireless signal booster device.

Previous research has made significant contributions to the localization and clustering of UAV applications. Companies like Facebook, Google, and Amazon are exploring the use of drones for product and data transportation along predetermined routes. Cellular UAV networks are also being investigated for addressing and public transit applications. Notable prototypes include the Nokia F-Cell project and the Huawei Wireless UAV lab under the Digital-Sky project in Shanghai, China.

However, there are still gaps in these studies, including high costs, limited optimization methods for identifying target nodes, and a lack of substantial self-configuration to coordinate drones using a centralized database that may be vulnerable to attacks. Recent papers have proposed novel algorithms for optimizing positioning. For example, the convolutional neural network (CNN) has been used for disaster relief in smart cities, and the Hough transform process and UVDAR system have been applied to the ultra-violet spectrum for improved efficiency.
Other experiments have explored distance-based docking, blurry approaches, radar-based high-precision 3D location, and using UAV relative distances for landing in restricted environments. Crowd surveillance systems employing multiple UAVs for monitoring walking groups have also been proposed. Security issues related to the Internet of Drones Things (IoDT) have been addressed using lightweight blockchain solutions, such as the combination of cuckoo and Hampel filters.

Additionally, there is research focusing on UAV applications in the context of 5G and 5GB wireless communication, including air-to-ground common channels and energy-efficient channels. UAVs have also been utilized for tasks such as search and rescue, trajectory planning, and artificial intelligence. Some studies have investigated the combination of UAVs and blockchain technology. Digital twins have been explored

for efficient delivery of medical resources during epidemics, utilizing deep learning algorithms for information prediction.

Various applications of UAVs have been studied, including 3D terrain modeling, joint UAV position optimization, cloud-edge computing, crowd systems, and safety-critical systems. Efforts have been made to improve internal UAV capabilities, such as optimizing hovering efficiency and energy-aware relay. Photogrammetry with multiple cameras onboard UAVs and environmental sense optimization are also areas of interest.

However, it is important to note that much of the existing work assumes fixed or non-modulated environments and lacks consideration for early weather prediction. Additionally, the decentralization of data and the vulnerability of fixed IoT data backups to infiltration have been overlooked in many studies. Table 1 provides an overview of the existing work limitations.

The previous studies along with their limitations :

- High Cost
- No creative form of optimization to identify the target node.
- No substantial self-configuration to coordinate drones using a
- centralized database that may be under attack.
- Efficiency.
- QoS.

In 2022, a research team in China published an article titled "Efficient and Secure WiFi Signal Booster via Unmanned Aerial Vehicles WiFi Repeater Based on Intelligence-Based Localization Swarm and Blockchain." This study aimed to leverage UAVs equipped with smart thermal cameras to monitor crowds in a specific

area and enhance weak wireless signals in remote emergency areas by utilizing a wireless signal booster device.

The research team utilized UAVs equipped with thermal cameras and signal repeaters to identify individuals in a specific area and amplify the wireless signal within that area. The procedure for requesting help through UAVs involved identifying specific regions as gathering points for people in need of assistance. Vision processing technology employed by the UAV cameras allowed for the recognition of indicators indicating a request for help.

To optimize the localization and clustering of the UAVs, a swarm intelligence localization (SIL) algorithm was employed. This algorithm improved the efficiency and accuracy of the UAVs' positioning within the designated area.

In order to ensure secure data transmission, blockchain technology was utilized to decentralize the data. By leveraging blockchain, the research team enhanced the security and integrity of the transmitted data.

Overall, this study contributed to the effective utilization of UAVs equipped with smart thermal cameras for crowd monitoring and the enhancement of weak wireless signals in remote emergency areas. The combination of thermal camera technology, swarm intelligence localization, and blockchain-based data security played a crucial role in achieving these objectives. (Amran et al., 2022)

Chapter 3 - Methodology

After a comprehensive research, few designs for a device that is capable of amplifying mobile signals in areas with no connection were selected. There have been few developments in the field of signal boosting; throughout my research, I came across only a handful of academic institutions' and a few individuals' projects and researches.

3.1 Concept Design 1 - Circuit diagram of the Colpitts oscillator

In 2017, Salihu Aliyu has conducted a project on Development of a GSM Signal Booster for the 900 and 1800 MHz Bands using BA-ANN. This project is conducted by Federal University of Technology in Nigeria. This paper introduces a low-cost GSM signal booster solution to address the problem of weak signal reception. The proposed approach combines the Bat Algorithm (BA) and Artificial Neural Network (ANN) to optimize component selection and predict performance. The BA, inspired by bats' echolocation abilities, efficiently selects the optimal circuit components. Meanwhile, the ANN, known for its ability to model complex problems, predicts the selected components' performance. The research aims to enhance the quality of service (QoS) for wireless communications in Nigeria.

Below figure illustrates the block diagram of the booster circuit, which incorporates a combination of the 900MHz and 1800MHz bands using a 2-1 multiplexer, a power amplifier, and an RF transmitter/receiver circuit. The power amplifier plays a crucial role in amplifying the weak received signal, while the bandpass filter ensures that only the desired GSM signal frequency is selected. The RF circuitry facilitates the reception and retransmission of the GSM signal.

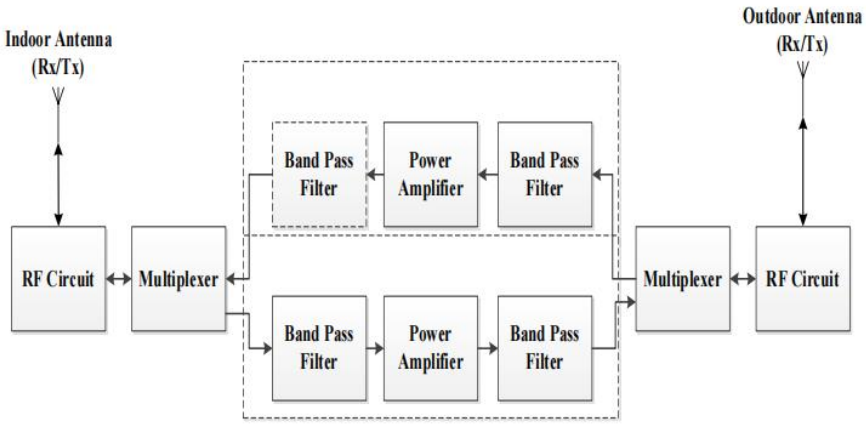


Figure 3 Block Diagram Concept 1

The GSM signal booster they proposed was specifically designed to enhance signals in the 900 and 1800 MHz bands, which are widely used by mobile networks in Nigeria. To accomplish this, they incorporated a bandpass filter that allows signals within these frequency ranges to pass through while blocking others. According to that paper, Nigerian network providers transmit signals to mobile devices within the frequency range of 935 to 960 MHz for the 900 MHz band and 1805 to 1880 MHz for the 1800 MHz band. Consequently, their bandpass filter for the 900 MHz range was designed to allow frequencies from 935 to 960 MHz, while the filter for the 1800 MHz range was tailored for frequencies from 1805 to 1880 MHz.

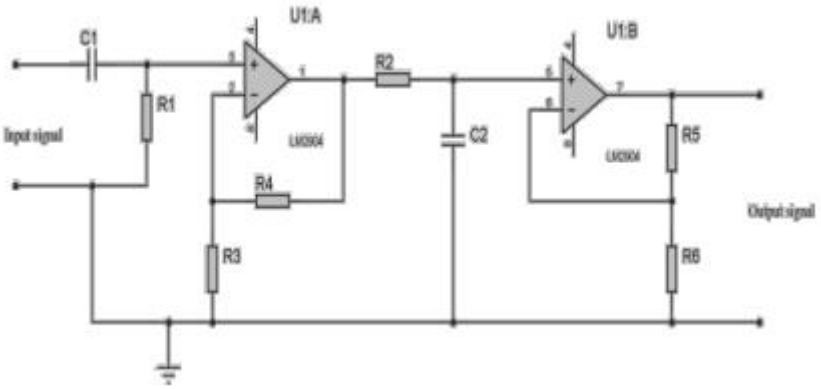


Figure 4 Circuit diagram of the bandpass filter

In this stage of the project, they employed an RF receiver and transmitter to handle the crucial tasks of receiving the weak signal and retransmitting the amplified signal. These circuits were meticulously designed, with a focus on integrating them within a wideband Colpitts oscillator framework. The selection of the specific resistors, R2 and R3, was accomplished through an innovative artificial intelligence technique utilizing the Bat algorithm. This metaheuristic algorithm draws inspiration from the echolocation abilities of bats and has shown promising results in various optimization problems.

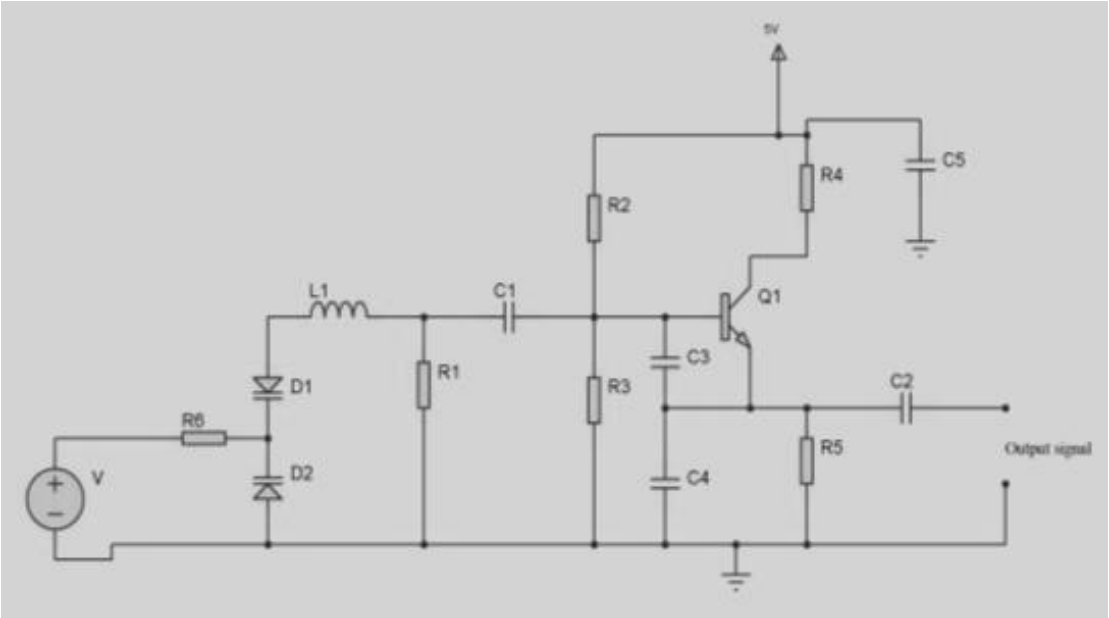


Figure 5 Circuit diagram of the Colpitts oscillator

The results of the experiments has showed a significant improvement in the weak signal, highlighting the usefulness of the GSM signal booster as a solution to the weak signal reception (WSR) problems experienced by users in both urban and rural regions. On the intended 2G network, the booster displayed an excellent maximum percentage increase of 73.59%. (Aliyu et al., 2017)

Limitations

- The booster won't be able to increase the signal in circumstances when it is so weak that the external antenna receives no signal at all.
- Limited to GSM Signals in range of 935 to 960 MHz and 1805 to 1880 MHz.
- Not suitable for a Drone.

3.2 Concept Design 2 - 4G Repeater Booster Device

Indonesia, Politeknik Negeri Sriwijaya university has designed a 4G repeater device in 2021. The paper discusses the design and configuration of a 4G repeater booster device operating at 1800MHz. The purpose of the device is to improve signal quality in mobile communication by amplifying the weak signal received by mobile devices due to obstacles or long distances from the base station. The repeater serves as a signal amplifier, extending the range of the signal transmission.

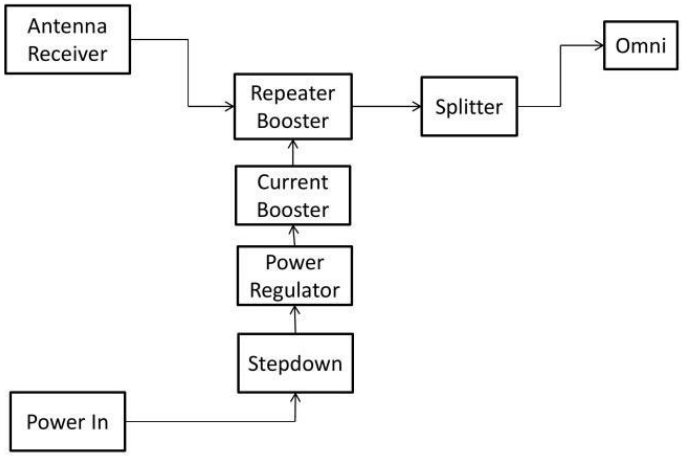


Figure 6 Hardware system Block Diagram

The difference between active and passive repeaters is explained. Active repeaters require an active power supply and components like amplifiers, while passive repeaters do not. In the case of 4G network signals, they may experience attenuation due to various factors such as air, weather, or obstructions. Therefore, a repeater is needed to enhance the transmitted signal for proper reception by users.

The paper emphasizes the importance of selecting suitable antennas for supporting the 4G network. The chosen antenna should be compatible, compact, and capable of operating over a wide frequency band. The 4G network operates at a frequency of 1800MHz and provides high-speed data transmission.

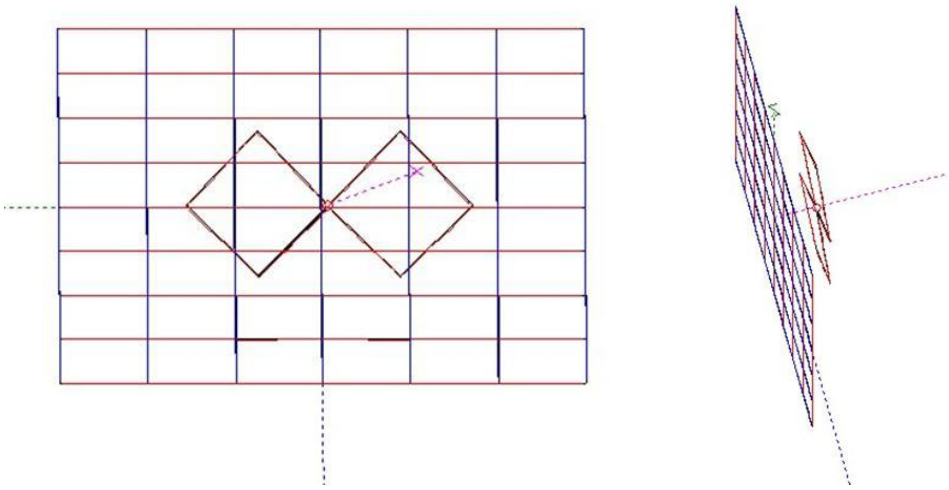


Figure 7 Biquad Antenna Shape and Reflector



Figure 8 Physical Form

The functions of network repeaters are discussed, including expanding signal coverage area, saving cost and time, and their applications in telephone, optical communication, and radio systems. The focus of this study is on 4G repeaters, which amplify the 4G signal using external omnidirectional antennas, signal amplifiers, and rebroadcast antennas. The amplified signal significantly increases the signal strength, benefiting both commercial and home users.

The paper also mentions the evolution of mobile network technologies from 2G to 4G LTE. LTE (Long-Term Evolution) is the latest standard in cellular network technology, designed to increase capacity and speed. The research methodology

involves the design steps, including electronic and mechanical design, component selection, and prototype development.

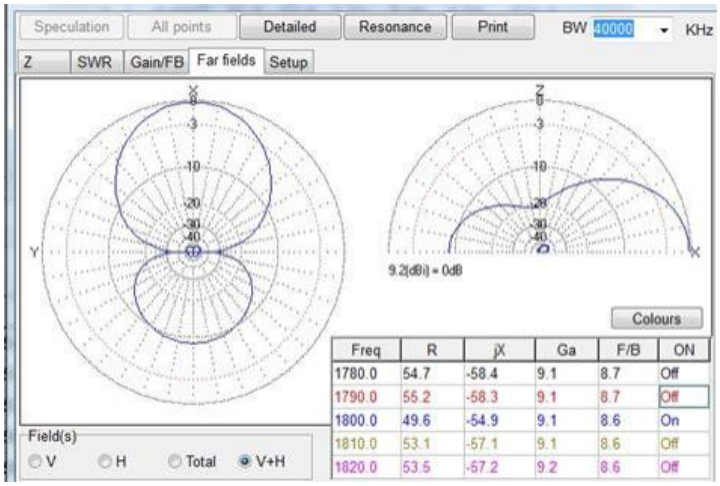


Figure 9 Resulting Impedance

In summary, this paper presents the design and configuration of a 4G repeater booster device operating at 1800MHz. It highlights the importance of signal amplification for improving mobile communication quality and discusses the functions and applications of repeaters in various communication systems. (Soim et al., 2021)

Limitations

- Focuses on only 4G Signal
- Not suitable for a drone

3.3 Concept Design 3 - Booster For LTE Network

The Mini Signal Booster for LTE Network is a project aimed at enhancing cellular signal strength in areas with poor network coverage. The objective of this study, conducted by researchers from the Electronics and Communication Engineering Department at the College of Science and Technology, Royal University of Bhutan,

was to design a mini-antenna system capable of amplifying the LTE signal for improved network connectivity.

The proposed system consists of a high-gain outdoor Yagi Uda antenna to receive the LTE signal at 1800 MHz and a patch panel indoor antenna. The weak signal received by the high-gain antenna is then fed into a directional signal booster system, where it undergoes amplification to increase its strength and power. The amplified signal is then retransmitted by the patch panel antenna in an omnidirectional radiation pattern, thereby boosting the network coverage in the targeted area.

The project methodology involved a qualitative research approach, relying on existing literature related to signal boosters and antenna design. The research team reviewed papers that discussed topics such as signal boosters for 4G networks, Yagi-Uda antenna design, wideband micro-strip antenna design, and GSM/DCS patch antenna design.

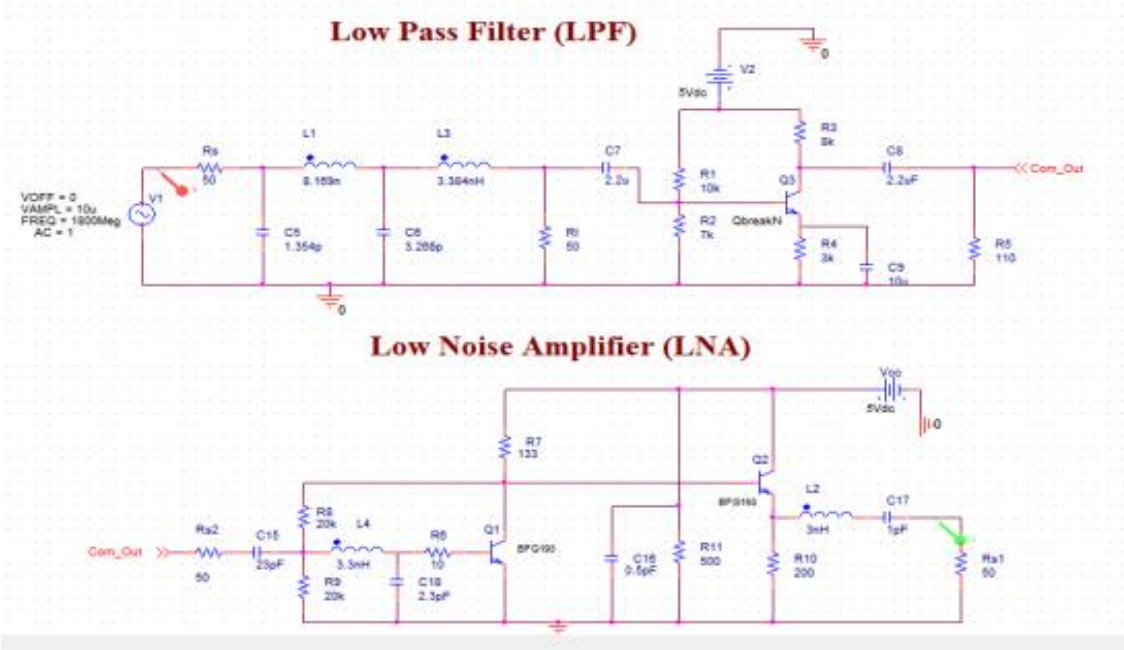


Figure 10 Low Filter and Low Noice Amplifier Circuit Diagram

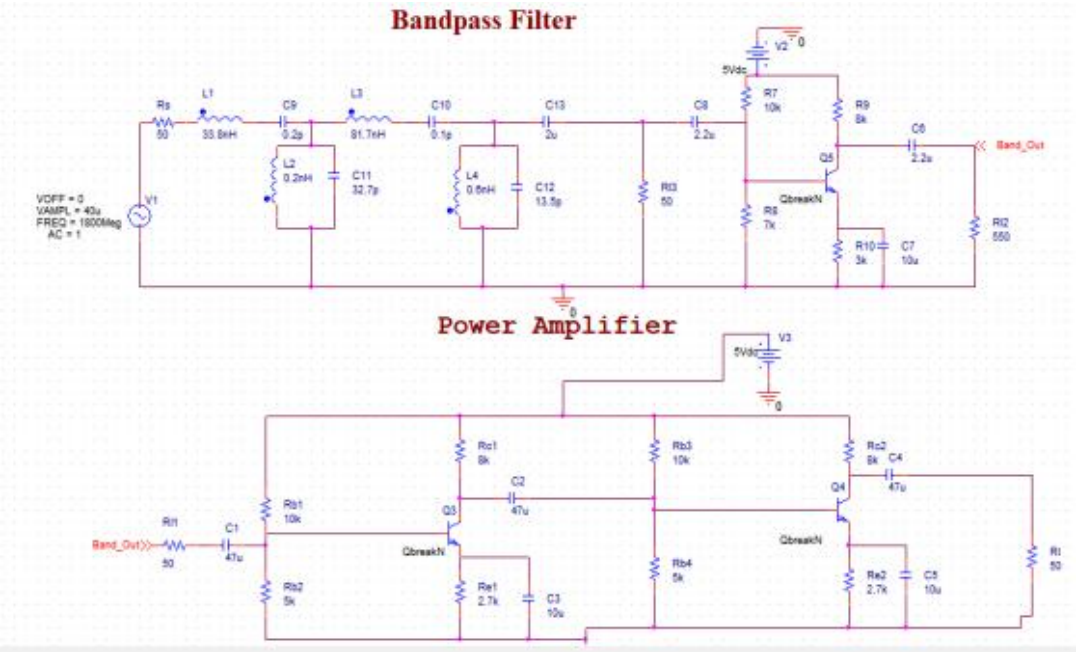


Figure 11 Bandpass Filter and Power Amplifier

The design process utilized software tools such as CST STUDIO SUITE and OrCAD Capture CIS Lite. The antenna system, including the Yagi-Uda antenna, low-pass filter, bandpass filter, low noise amplifier (LNA), and power amplifier, was individually designed and integrated using simulation circuits for testing and verification.

Simulation results demonstrated the effectiveness of the designed system. The Yagi-Uda antenna exhibited high directivity and gain, making it suitable for capturing weak signals. The integrated low-pass filter and LNA successfully amplified the received signal, while the bandpass filter and power amplifier further enhanced the signal quality. The patch antenna, based on microstrip technology, achieved omnidirectional radiation and wider beam coverage.

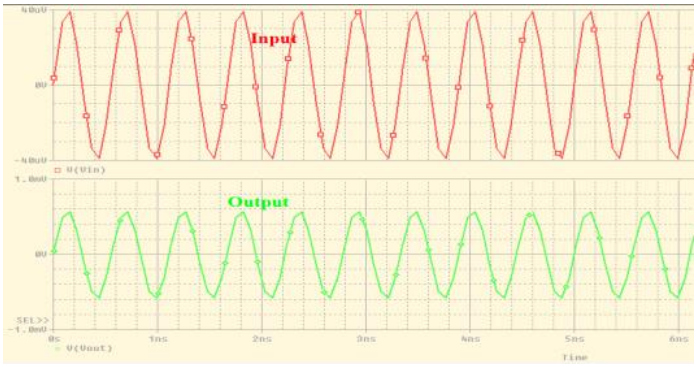


Figure 12 Input Output Wavefrom

In conclusion, the Mini Signal Booster for LTE Network project presents a mini-antenna system capable of amplifying the LTE signal to improve network connectivity in areas with poor coverage. The design and integration of components, including the Yagi-Uda antenna, filters, and amplifiers, were successfully implemented using simulation software. The proposed system offers a promising solution to enhance cellular network performance and address the challenge of weak signal coverage. (Melong et al., 2021)

LImitations

- Focuses on only LTE Signal
- Not suitable for a drone

3.4 Concept Design 4 - RF Power Amplifier and Improved Linearity with DPD

While Solving issues faced during the MATLAB simulation,a method was found in Modeling of RF (Radio Frequency) power amplifiers and the implementation of Digital Pre-Distortion (DPD) techniques to enhance the linearity of wireless transmitters. RF power amplifiers are essential components in wireless communication systems, responsible for boosting the power of the transmitted signals. However, these amplifiers can introduce nonlinearities, leading to distortion and degradation of signal quality. To mitigate these issues, DPD techniques are employed to compensate for the nonlinear behavior of power amplifiers and improve overall transmitter linearity. This section explains how the concepts of RF power amplifier modeling and Digital Pre-Distortion (DPD) techniques can contribute to mobile signal boosting. By applying these techniques, the aim is to improve signal quality, extend coverage, and enhance overall performance in mobile communication systems.

3.4.1 RF Power Amplifier Modeling:

Nonlinear characteristics that have a profound effect on performance. They had examine and describe gain compression, a nonlinear process that causes lower gain at high input power levels. Another crucial nonlinear phenomena known as intermodulation distortion is carefully explored in light of its detrimental effects on signal quality. To fully capture and reflect the complicated behavior of power amplifiers, exact mathematical models are created. A variety of modeling methods were looked exploring to handle the nonlinearities. Memoryless models, which ignore memory effects, offer an intuitive yet accurate illustration of power amplifier performance. On the other hand, memory effects are included in memory polynomial models, allowing for more precise definition.

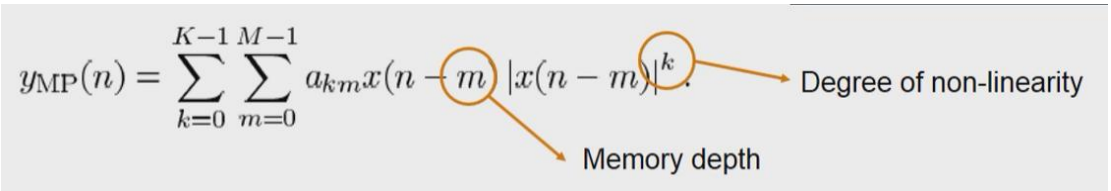


Figure 13 Fitting Memory polynomial

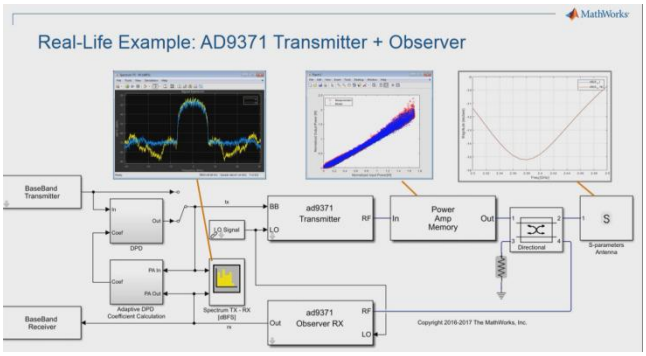


Figure 14 Transmitter and Observer

3.4.2 Digital Pre-Distortion (DPD):

Digital Pre-Distortion (DPD), a technique widely used to mitigate the nonlinear effects of RF pow er amplifiers. DPD operates by generating an inverse distortion signal that cancels out the nonlinear distortions introduced by the amplifier. This discusses different DPD algorithms, such as the Look-Up Table (LUT) based approach, Volterra series based approach, and adaptive algorithms like the Least Mean Squares (LMS) and Recursive Least Squares (RLS). The advantages, challenges, and trade-offs associated with each DPD method are examined.

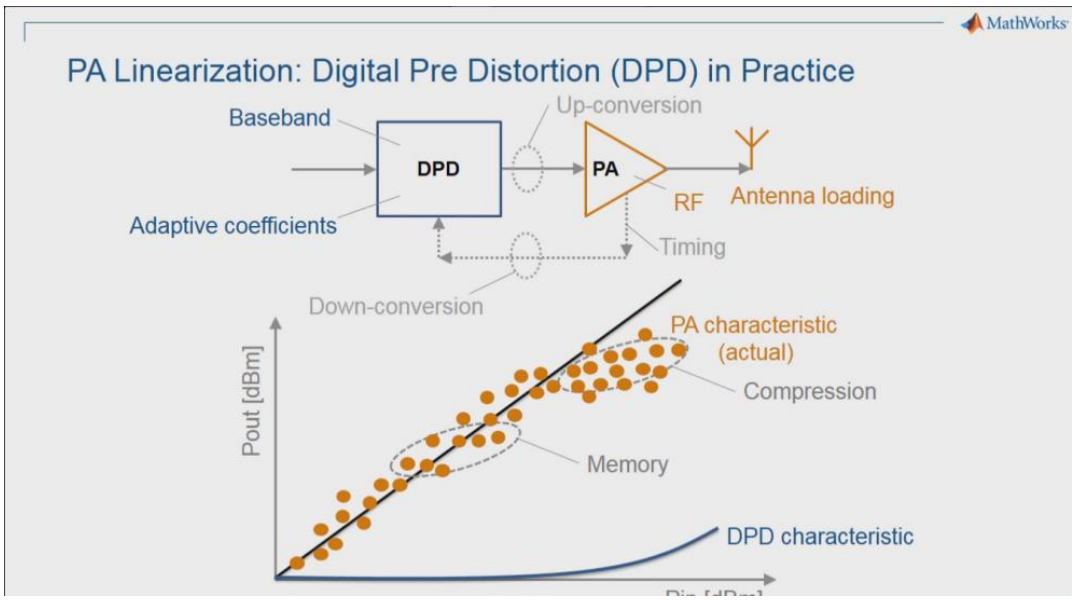


Figure 15 Digital Pre-Distortion

3.4.2 Implementing DPD in Wireless Transmitters:

On the explanation of the practical implementation of DPD in wireless transmitters, It covers the system architecture for incorporating DPD into a transmitter chain and the necessary signal processing blocks involved. Key considerations, such as DPD training and adaptation, power consumption, latency, and real-time processing requirements, are discussed. Additionally, the document highlights the importance of parameter estimation and adaptation algorithms to ensure the DPD system's effectiveness over varying operating conditions.

By deploying such systems in mobile networks, signal boosters equipped with RF power amplifier modeling and DPD techniques can effectively enhance signal strength, extend coverage, and provide a more reliable mobile communication experience. The improved linearity and reduced distortions result in stronger and cleaner signals, enabling better reception in areas with weak signal coverage or interference. Mobile users can enjoy improved call quality, faster data speeds, and a more seamless mobile communication experience overall.

The theoretical plan focuses on boosting the mobile signal by following a systematic approach. It begins with modeling RF power amplifiers to analyze their nonlinear characteristics, including gain compression and intermodulation distortion. Mathematical models are developed to accurately represent the amplifier behavior, paving the way for optimization and distortion reduction.

Digital Pre-Distortion (DPD) techniques will be employed to mitigate distortions caused by power amplifiers. Customized DPD algorithms will be implemented to counteract nonlinearities and enhance linearity through signal processing and feedback mechanisms. This step aims to pre-compensate for distortions, resulting in improved signal quality and reduced degradation.

Following DPD implementation, the components will be integrated into a mobile signal boosting system, and comprehensive testing will be conducted. This testing

phase will evaluate the system's performance in terms of signal strength, coverage, and overall efficiency.

Based on test results, the parameters of RF power amplifier models and DPD algorithms will be optimized to achieve the best linearity, efficiency, and signal quality. This iterative process ensures continuous enhancement of the mobile signal boosting capabilities.

The optimized mobile signal boosting system will be deployed in real-world scenarios to assess its performance and effectiveness. Evaluations will be conducted in diverse environments and usage scenarios, collecting user feedback and performance metrics to validate the success of the theoretical plan.

By following this theoretical plan, the aim is to enhance mobile signal quality and reliability. Through RF power amplifier modeling and DPD techniques, distortions will be compensated, resulting in improved linearity, increased signal strength, and enhanced mobile signal reception. Ultimately, this plan aims to provide users with a superior and consistent mobile communication experience.

By following this theoretical plan, we aim to boost mobile signal quality and reliability. The combination of RF power amplifier modeling and DPD techniques will allow us to accurately compensate for distortions, resulting in improved linearity, increased signal strength, and enhanced mobile signal reception. Ultimately, this plan will contribute to providing users with a better and more consistent mobile communication experience.

Incorporating RF power amplifier modeling and DPD techniques into mobile signal boosting systems offers significant benefits in terms of signal quality, coverage extension, and overall performance enhancement. By accurately modeling power amplifiers and compensating for their nonlinear behavior, signal degradation and distortions can be minimized. This leads to improved signal strength, reduced dropped calls, and enhanced mobile communication experiences for users. By leveraging advanced techniques and implementing them in practical mobile network

infrastructures, the goal of seamless and reliable mobile signal boosting can be achieved.

3.5 Concept Design 5 - Crystal Oscillator based Mobile Signal Booster

While research period I came across few methods of signal boosting using crystal oscillators. In this study, various methods for boosting mobile signals using a crystal oscillator were investigated. The aim was to select an effective and practical approach for improving mobile network coverage. Three methods were explored, all sharing a similar concept: receiving the signal with a receiver antenna, amplifying it using a crystal oscillator, and transmitting the boosted signal through a transmitter antenna. These methods were evaluated based on their performance, feasibility, and compatibility with the drone platform.

The system at hand comprises three essential components: a receiver antenna, a crystal oscillator-based amplifier circuit, and a transmitter antenna. The receiver antenna plays a crucial role by capturing faint mobile signals in the vicinity, necessitating its design to align precisely with the desired frequency bands of the target mobile network.

At the heart of the system lies the crystal oscillator-based amplifier circuit, representing its core functionality. This circuit undertakes the task of amplifying the received signal obtained from the receiver antenna. Employing the power of a crystal oscillator, it generates a stable frequency that serves as the bedrock for signal amplification. Key components within the circuit include the crystal oscillator itself, as well as carefully selected resistors, capacitors, and transistors. These elements work in unison to fine-tune, filter, and magnify the signals, with the aim of providing ample gain to enhance weak mobile signals while upholding signal integrity and mitigating distortion.

The transmitter antenna assumes the responsibility of rebroadcasting the amplified signal into the surrounding environment. Much like its counterpart, the receiver antenna, it should be meticulously designed to operate within the specified frequency

bands. Additionally, it should possess a high gain characteristic to ensure the efficient transmission of the boosted signal.

By seamlessly integrating these three components, the system empowers us to overcome the challenges posed by feeble mobile signal reception. With a blend of technical prowess and a human-centric approach, it equips us with the capability to amplify and propagate signals, revolutionizing the way we connect with mobile networks.

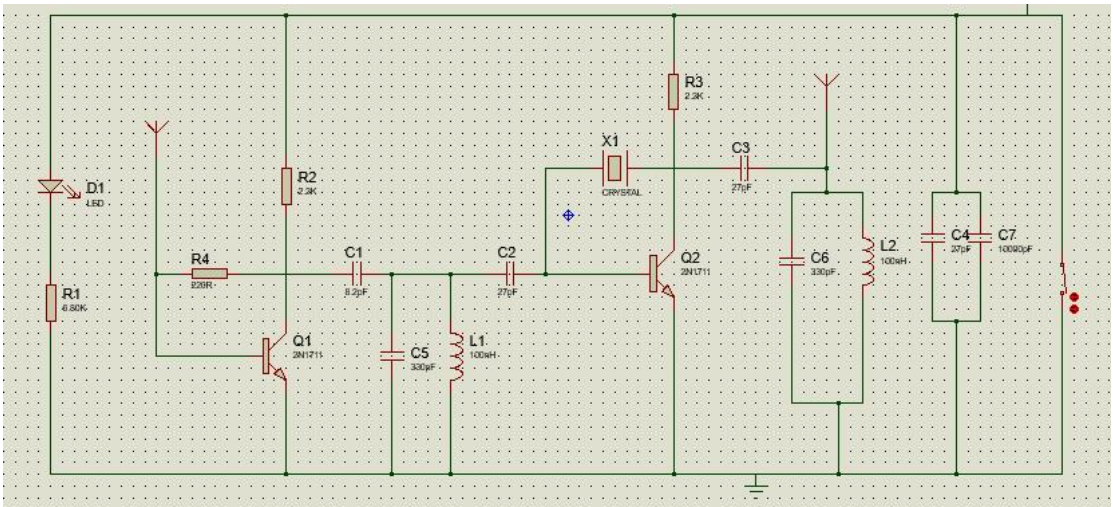


Figure 16 Circuit Diagram

While the exact range specifications for the mentioned components have not been specified, it is reasonable to assume that upgrading the receiver antenna, crystal oscillator-based amplifier circuit, and transmitter antenna can potentially enhance the system's range and effectiveness.

To boost the range, the receiver antenna can be optimized by using antennas with higher gain and implementing advanced signal capture techniques such as beamforming or adaptive filtering. These improvements can enable the antenna to detect weak mobile signals over greater distances and improve signal quality by reducing interference.

Upgrading the crystal oscillator-based amplifier circuit involves using superior components, refining the circuit design, and implementing advanced amplification

techniques. By employing high-quality crystal oscillators with improved stability and lower phase noise, combined with carefully selected resistors, capacitors, and transistors, the circuit can amplify weak signals with higher gain, reduced distortion, and improved linearity.

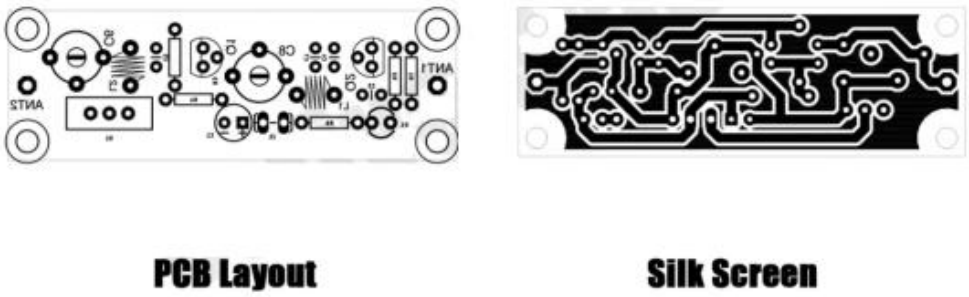


Figure 17 PCB Layout and Silk Screen

Similarly, optimizing the transmitter antenna involves selecting antennas with higher gain and directional characteristics, as well as utilizing advanced transmission technologies like MIMO or beamforming. These enhancements can increase the radiated power and focus the transmitted signal in specific directions, thereby extending the range and improving signal coverage and penetration through obstacles.

While the precise impact of these component upgrades on the system's range and effectiveness would require empirical testing, it is reasonable to expect that such improvements would result in overall advancements, allowing for better signal reception over greater distances and improved performance in various environments.

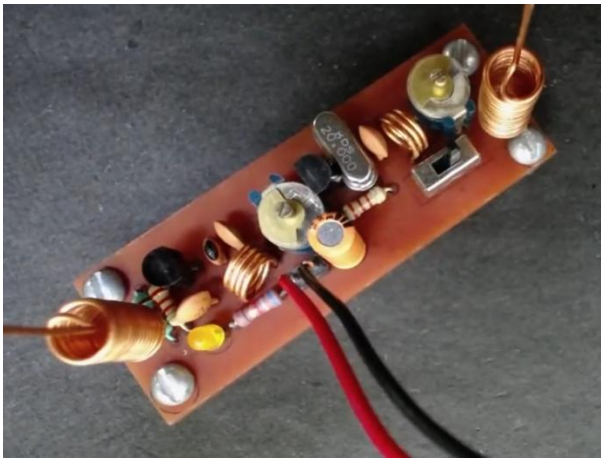


Figure 18 Hardware Implementation

3.6 Concept Design Drone

The research paper mentioned above titled "Efficient and Secure WiFi Signal Booster via Unmanned Aerial Vehicles WiFi Repeater Based on Intelligence Based Localization Swarm and Blockchain" explains the theoretical aspects of using UAVs to enhance wireless signals in isolated cities. The study proposes the utilization of swarm intelligence localization (SIL) in conjunction with blockchain technology to achieve synchronized, secure, and efficient wireless signal boosting.

SIL is an intelligent optimization algorithm inspired by the behavior of animal swarms, such as bird colonies. It aims to determine the optimal position of UAVs based on the direction and location information of neighboring particles. In this context, follower UAVs collaborate to solve the localization problem by measuring their local positions using the best private position (Pbest) and the best swarm position (Gbest). Through information exchange, the follower UAVs collectively choose the best overall position for the leader UAV. Each UAV retains its best position and velocity values after each iteration round, with coordination among swarm members being key to achieving the desired level of intelligence. The computational complexity of the swarm is represented by three follower UAV candidates (Xi(t)) and their corresponding velocities (Vi(t)). Each candidate maintains its own best location (Pi(t)), while the swarm as a whole shares the best position (G(t)). By updating their positions and velocities (Vi(t + 1) and Xi(t + 1)) based on Pi(t), G(t), and other elements, the follower UAVs navigate to new positions within the search space.

The paper assumes several factors, including the use of UAVs for wireless signal boosting in remote cities during emergency situations, SIL being an effective solution for localization, blockchain technology offering decentralized data sharing advantages, and the synergistic operation of SIL and blockchain for synchronized signal boosting.

Overall, the study presents the theoretical framework of utilizing UAVs, SIL, and blockchain to enhance wireless signals in isolated cities, with a focus on achieving efficient and secure signal boosting.

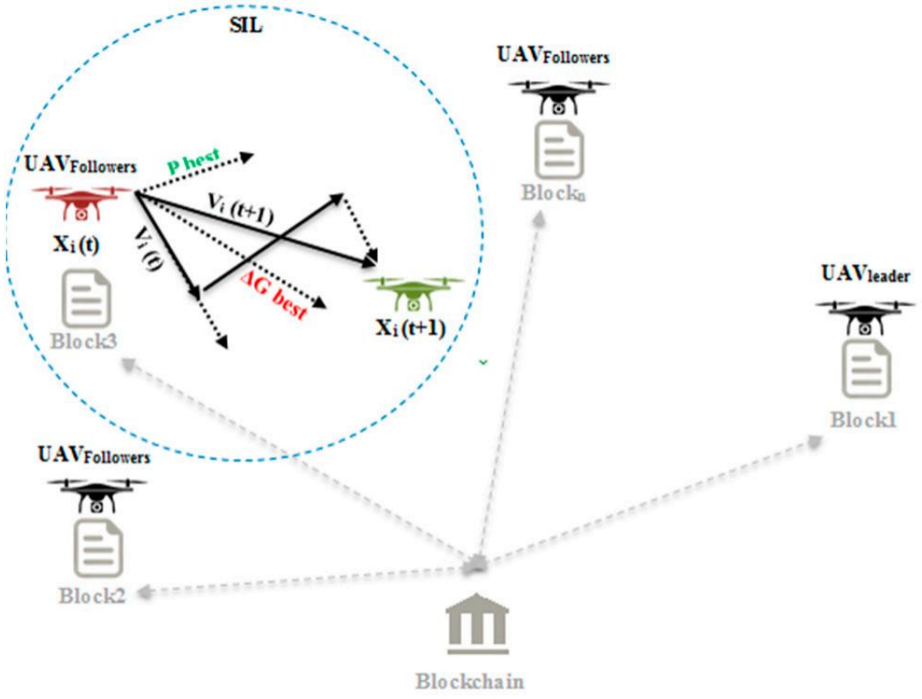


Figure 19 UAV Swarm Intelligence

After delving deep into this research paper and further exploring related studies, I gained profound insights into the specifications and qualities that my drone should embody. The comprehensive analysis provided me with a clear vision of the essential features necessary for optimal signal boosting performance.

The Signal Boosting Drone is a specialized aerial vehicle designed to provide signal boosting capabilities in areas with weak or unstable wireless signals. This drone operates manually, leveraging advanced technologies to enhance signal strength and improve wireless communication. With its signal amplification equipment and intelligent signal analysis, the drone ensures reliable connectivity and seamless communication for users in need.

Key Features:

- Signal Amplification Equipment: The Signal Boosting Drone is equipped with high-gain antennas and signal amplifiers specifically designed to boost wireless signals. These components allow the drone to extend the coverage area and enhance the quality of communication signals.

- Advanced Operation: Using onboard sensors, GPS, and advanced flight control systems.

- Stable and reliable communication system to transmit boosted signals back to the ground station.

- Real-time Signal Analysis: The drone incorporates advanced signal processing techniques to analyze the quality and strength of wireless signals in real-time. It continuously monitors signal parameters such as signal-to-noise ratio, channel interference, and signal stability to make informed decisions for signal optimization.

- Adaptive Signal Boosting: Based on real-time signal analysis, the drone dynamically adjusts its signal amplification parameters to optimize signal quality. It can adaptively change the amplification gain, beamforming direction, or frequency allocation to maximize signal coverage and minimize interference, ensuring a reliable and stable connection.

- Compact and Lightweight Design: The drone features a compact and lightweight design, enabling easy transportation and deployment. It is equipped with a robust propulsion system for stable flight, allowing access to remote or hard-to-reach areas where signal coverage is limited.

The Signal Boosting Drone concept offers an innovative solution to enhance wireless signals in areas with weak or unstable connectivity. With its signal amplification equipment, autonomous operation, real-time signal analysis, and adaptive signal boosting capabilities, the drone effectively improves signal strength and coverage. Whether deployed in emergency scenarios or remote locations, this drone provides reliable wireless connectivity, ensuring seamless communication for users in need.

Chapter 4 Design and Implementation

In this chapter, two methods of mobile signal boosting are thoroughly discussed. The first method involves utilizing a crystal oscillator as an amplifier, focusing on the design aspects and software implementation required for effective signal boosting. The second method explores the modeling of RF Power Amplifiers and the implementation of Digital Pre-Distortion (DPD) techniques to enhance linearity and improve signal quality. Both methods offer distinct approaches to mobile signal boosting, combining hardware and software implementations to optimize signal strength and overall performance. This chapter provides a comprehensive understanding of the design considerations and implementation strategies for achieving efficient mobile signal boosting.

4.1 Basic Architecture

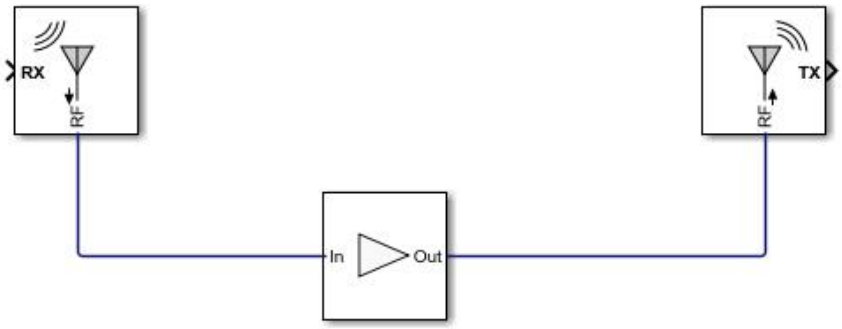


Figure 20 Basic Design

Both the methods share a common fundamental approach to mobile signal boosting. They both involve a two-step process of receiving the signal and then amplifying it before transmitting the boosted signal. While the first method utilizes a crystal oscillator as an amplifier, emphasizing design and software implementation, the second method focuses on modeling RF Power Amplifiers and implementing DPD

techniques for improved linearity. Despite their distinct approaches, both methods aim to optimize signal strength and enhance overall performance. By receiving the signal, amplifying it, and transmitting the boosted signal, these methods harness the power of technology to ensure efficient and effective mobile signal boosting.

4.2 Method 1 - Mobile Signal Booster using Crystal Oscillator

4.2.1 Crystal Oscillator

A suitable frequency source for the circuit is provided by the use of a 16Hz crystal oscillator in this project. Timing within the circuit is constant and accurate mainly to the crystal oscillator's operation at a specified resonance frequency of 16Hz. It performs the function of an accurate clock signal generator, allowing synchronization and coordination of numerous circuit components. The mobile signal booster design's special needs, taking into account elements like signal amplification, filtering, and transmission, lead to the selection of the 16Hz frequency. The crystal oscillator's inherent stability and low phase noise qualities make it a good choice for ensuring the circuit performs consistently and dependably, which helps with the circuit's capacity to boost and amplify signals effectively.

4.2.2 Antenna

Due to its unique design features, the antenna made with 15 turns of wire measuring 5 mm in diameter and 20 AWG is regarded suitable for the circuit. The 15 turns provides the optimum number of windings for the effective transmission and collection of electromagnetic signals. A acceptable size while yet being able to handle the appropriate frequency range effectively, the 5mm diameter ensures a balance between compactness and performance. Choosing wire that is 20 AWG provides proper electrical conductivity and reduces losses inside the antenna. This antenna is well-suited for the signal boosting needs of the circuit since it has the turns, diameter, and wire gauge needed to successfully receive and transmit signals within the desired frequency range.

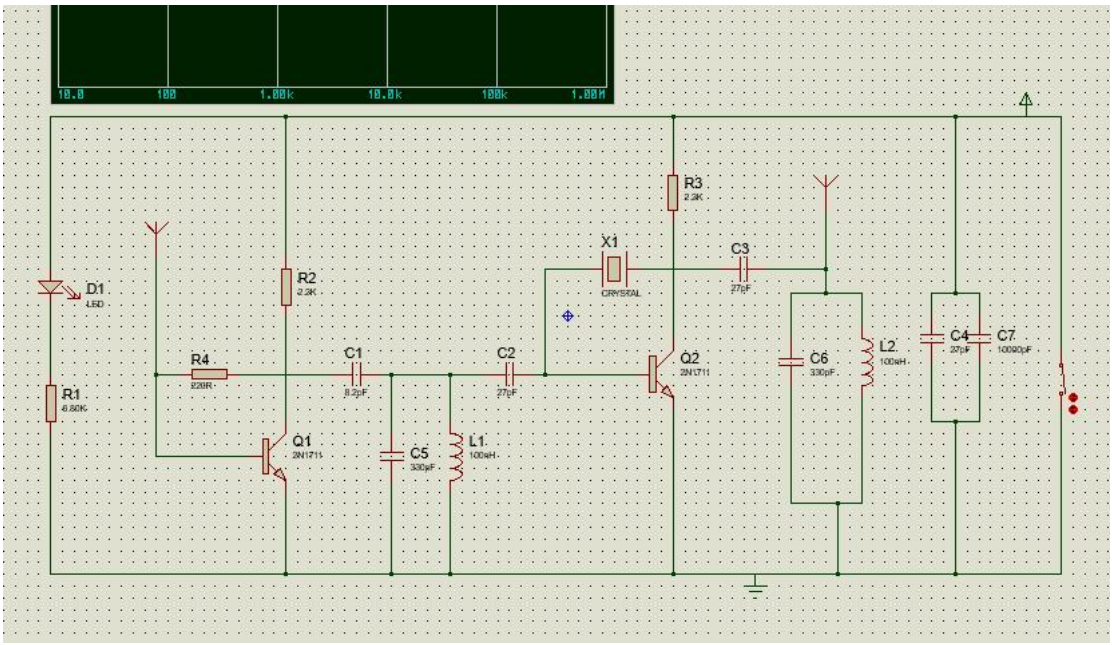


Figure 21 Proteus Diagram

The completion of the circuit and the implementation of effective signal filtering before and after the crystal oscillator rely on the precise characteristics and performance of capacitors, resistors, and transistors. Capacitors are essential components in the circuit, functioning as DC blockers and AC signal conduits. Through energy storage capabilities, capacitors selectively allow AC signals to propagate while blocking unwanted DC components. The appropriate selection of capacitance values determines the circuit's frequency response and filtering characteristics, ensuring optimal signal transmission.

Resistors play a vital role in regulating current flow within the circuit. They serve multiple purposes, including signal amplitude determination, voltage level setting, and biasing functions. By carefully selecting resistor values, the gain and impedance of the circuit can be fine-tuned to match the specific requirements of the signal amplification process. Additionally, resistors contribute to signal attenuation and facilitate proper impedance matching between different circuit components, enabling efficient signal transfer.

Transistors, as active semiconductor devices, are crucial for signal amplification and modulation in the circuit. They serve as both amplifiers and switches, amplifying weak input signals to desired levels and controlling signal flow through the circuit. By precisely managing the biasing and operating conditions of the transistors, the amplified signal can be accurately modulated and shaped according to specific requirements.

To summarize, the incorporation of capacitors, resistors, and transistors in the circuit design ensures effective signal filtering, impedance matching, gain control, and modulation. These components collectively enable optimal performance and functionality of the mobile signal booster, facilitating enhanced signal reception and transmission.

The origin and the foundation of this crystal oscillator based mobile signal booster lacked the accuracy of the circuit and the underlying methodology. Which raised the question about it's reliability and accuracy. After Researching and adjusting modifications, the intended purposes of the components used in the circuit was assumed.

4.2.3 Hardware Implementation

For the hardware implementation of the aforementioned circuit, a comprehensive collection of components was acquired to ensure the successful construction of the mobile signal booster. The resistor selection included 2.2K, 200K, and 6.8K resistors, each serving specific purposes within the circuit. These resistors play a crucial role in controlling the current flow, determining voltage levels, and establishing biasing conditions, which are essential for achieving the desired amplification and filtering characteristics.

In addition to resistors, capacitors were carefully chosen to fulfill their distinct functions within the circuit. The inclusion of 1µF, 27pF, and 8pF capacitors enabled effective signal filtering and impedance matching. The 1µF capacitor blocks DC components, allowing only AC signals to pass through, while the smaller capacitance

values, such as 27pF and 8pF, contribute to precise frequency response shaping and suppression of undesirable frequencies. These capacitors act as energy storage elements, maintaining signal integrity and enhancing the overall performance of the mobile signal booster.

The antenna configuration is a critical aspect of the hardware implementation. The primary antenna consists of 15 turns of 5mm diameter wire with 20 AWG, providing the necessary inductance and impedance characteristics for signal reception and transmission. This carefully designed antenna design ensures optimal signal capturing and propagation efficiency. Additionally, a secondary antenna comprising 4 turns of 5mm diameter wire with 20 AWG is employed to further enhance signal reception in specific scenarios, thereby augmenting the overall boosting capability of the system.

In order to facilitate signal amplification, a crucial component employed in the circuit is a 16Hz crystal oscillator. This crystal oscillator serves as a fundamental element, generating a highly accurate and stable frequency that serves as the foundation for the signal amplification process. By providing a precise frequency reference, the crystal oscillator plays a pivotal role in maintaining signal integrity and preventing any distortion during the amplification stage. The inclusion of a crystal oscillator in this hardware implementation ensures the mobile signal booster's performance remains reliable and consistent, bolstering its effectiveness in enhancing signal quality and strength.

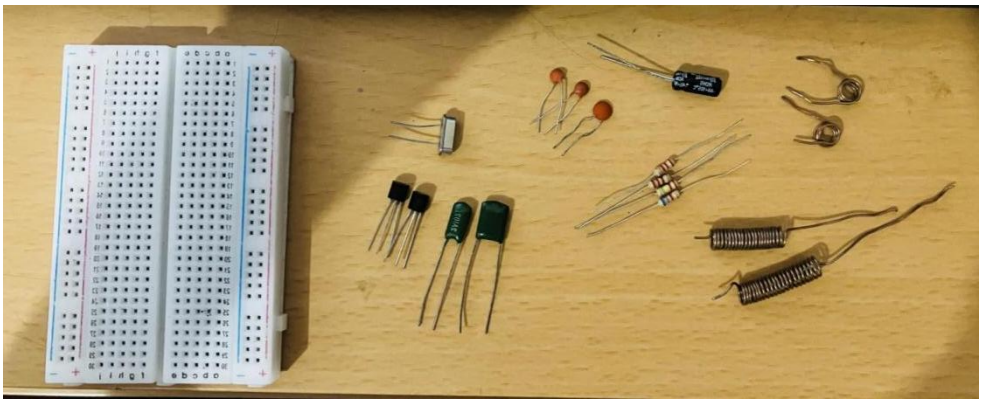


Figure 22 Hardware Components - Crystal Oscillator, antenna,capacitors,resistors

The hardware implementation of the circuit was carried out on a breadboard, allowing for easy modification and experimentation. This setup facilitates quick adjustments and alterations, ensuring that the circuit's performance can be fine-tuned and optimized based on empirical observations and measurements. The breadboard configuration also enables convenient testing of different component values and arrangements, enhancing the flexibility and versatility of the hardware implementation process.

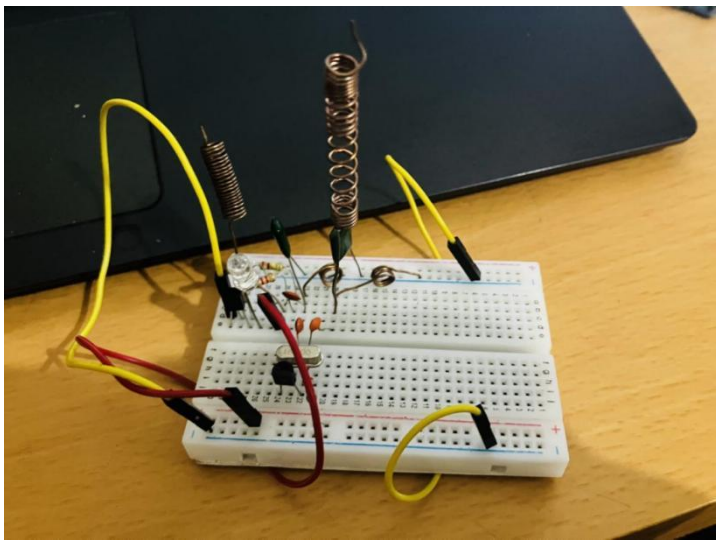


Figure 23 Hardware Model - Crystal Oscillator, antenna,capacitors,resistors

In conclusion, the hardware implementation of the mobile signal booster involves meticulous component selection, including resistors, capacitors, a crystal oscillator, and antennas, along with the utilization of a breadboard for ease of modification and testing. The combination of these elements, tailored to their specific roles within the circuit, enables the efficient boosting of mobile signals and enhances the overall performance of the signal booster system.

4.3 Method 2 - RF Power Amplifier and Improved Linearity with DPD to boost signal

The purpose of this method is to enhance the performance of mobile signals by focusing on two key aspects: modeling RF power amplifiers and increasing wireless transmitter linearity through the use of Digital Pre-Distortion (DPD) techniques. By employing modeling techniques, the characteristics of RF power amplifiers can be accurately captured and understood. This enables the identification and compensation of non-linearities and distortions introduced by the amplifiers. By implementing DPD, the signal linearity can be improved, resulting in boosted mobile signals that exhibit higher quality, reduced interference, and increased coverage. Through the combination of RF power amplifier modeling and DPD, this method aims to optimize mobile signal transmission and enhance the overall user experience.

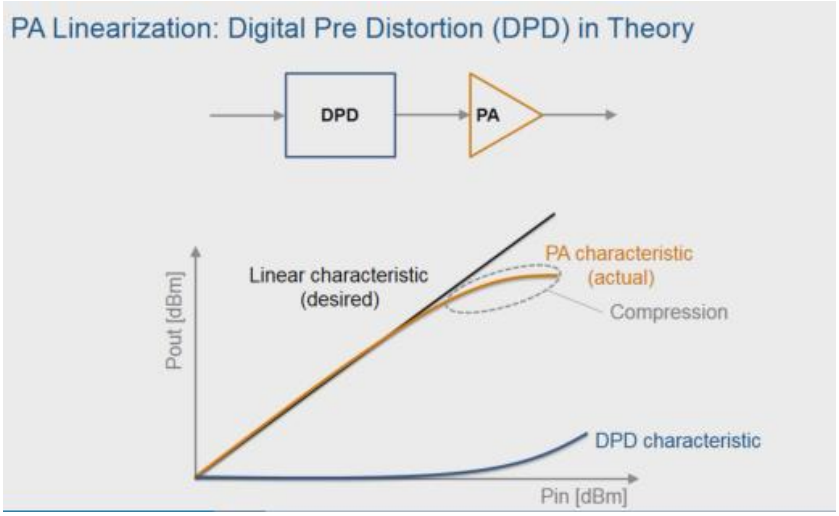


Figure 24 PA Linearization : DPD in theory

Designing RF transmitters has become increasingly challenging due to emerging standards such as 5G and the need for multistandard transmitters. These factors pose complex requirements in terms of frequency bands, modulation schemes, power levels, and linearity. The transition to 5G brings higher data rates, wider bandwidths, and more advanced modulation schemes, demanding transmitters with exceptional performance and flexibility.

In the case of multistandard transmitters, the challenge lies in supporting multiple wireless communication standards within a single device. This requires the integration of various RF components, such as power amplifiers, filters, and mixers, to accommodate the diverse frequency bands and modulation schemes of different standards. Achieving efficient coexistence and optimal performance across these standards adds to the complexity of transmitter design.

4.3.1 Digital Pre distortion to use PA linearity in theory

Based on Dr. Giorgia Zucchelli's explanation of Digital Pre distortion to use PA linearity, The principle behind improving the linearity of power amplifiers is relatively simple: the power amplifier itself introduces nonlinearity, while the desired characteristic is linearity. To achieve linearity, a digital algorithm called Digital Pre-Distortion (DPD) is introduced. Before the signal reaches the power amplifier, it is pre-distorted by the DPD algorithm. The DPD algorithm is designed to have characteristics that cancel out the nonlinearity introduced by the power amplifier.

However, the theory and implementation of DPD algorithms are more complex. implies that the power amplifier's output at a given moment depends on its input signals from the past. Power peak-to-average ratio, bandwidth, type of modulation, and other features of the input signal all have an impact on how the power amplifier behaves. To properly account for the impacts of the power amplifier, the DPD algorithm must be adaptive and run in a closed feedback loop configuration.

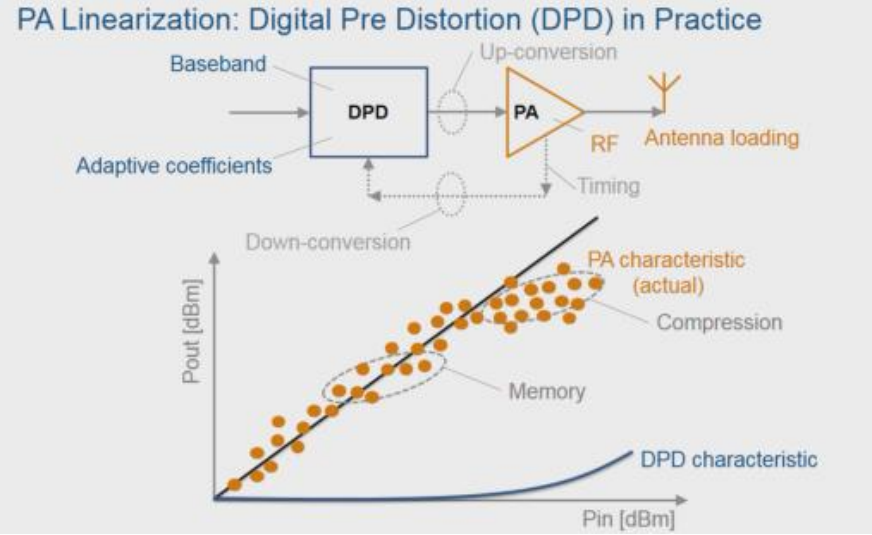


Figure 25 PA Linearlization

DPD algorithms are implemented in the baseband while power amplifiers and other RF components function at high frequencies, which adds to the complexity. In order to ensure correct functioning and synchronization between the baseband and RF components, up-conversion and down-conversion stages must be used.

Timing imperfections, such as synchronization issues, play a crucial role in the stability of the system. Neglecting these imperfections can lead to instability and degraded performance. Additionally, the loading effects of the antenna on the power amplifier (PA) and the strategy employed to adaptively compute the correcting coefficients are other important aspects that cannot be overlooked. These factors significantly impact the overall performance and effectiveness of the DPD algorithm.

To address this complex design problem and develop a DPD algorithm, it is necessary to accurately represent the PA model. The behavior of the PA is influenced by both linear characteristics and memory effects, which are determined by the input signal characteristics. PA models that capture these effects often employ the Volterra series, which represents a mathematical framework for describing nonlinear systems. However, the identification of these complex models typically involves intricate procedures.

4.3.2 Software Implementation

The software implementation of Digital Pre-Distortion (DPD) in MATLAB, guided by Dr. Giorgia Zucchelli's insights, played a pivotal role in harnessing power amplifier (PA) linearity to optimize system performance. By leveraging Dr. Zucchelli's technical explanation and harnessing the computational capabilities of MATLAB, the DPD algorithm was meticulously developed and seamlessly integrated into the system. This software-based DPD implementation effectively mitigated nonlinearities inherent in the PA, resulting in substantial improvements in signal quality and showcasing the practical application of advanced RF signal processing techniques. The involvement of Dr. Zucchelli, with her deep expertise in the field, significantly contributed to the advancement of RF signal processing methodologies in this software-driven implementation. Mobile signal boosting involves improving the quality and strength of the mobile signal to enhance communication performance. The method described, involving modeling RF power amplifiers and increasing wireless transmitter linearity with Digital Pre-Distortion (DPD), plays a crucial role in achieving this objective. Leveraging the capabilities of Simulink, engineers can simulate and evaluate the transmitter model. Simulink provides a powerful environment for exploring scenarios, assessing DPD strategies, and optimizing real-world transmitter behavior.

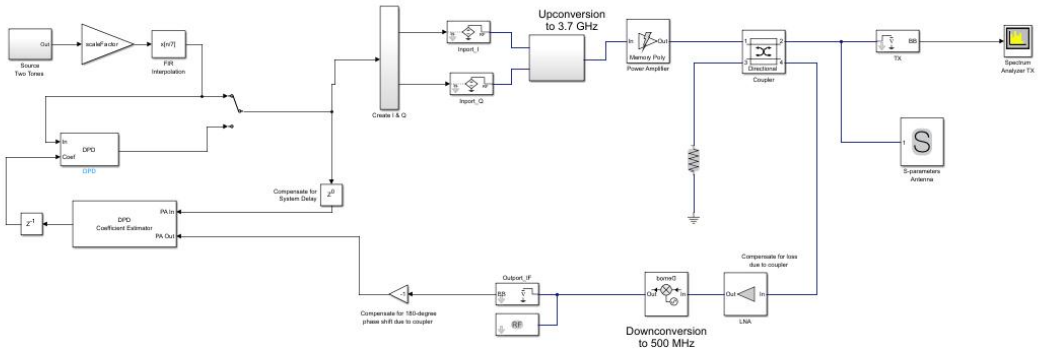


Figure 26 Simulation Configuration 1

The integration of RF blocks within the circuit facilitates accelerated RF simulations, accounting for nonlinearity, impedance mismatches, and learning imperfections. The transmitter model, a simplified behavioral representation, comprises several modification stages.

The IQ modulator and power amplifier (PA) form the initial components of the transmitter model. The IQ modulator facilitates baseband signal modulation onto a carrier frequency, while the PA amplifies the modulated signal. These components interface with an antenna described by S-parameters, enabling accurate characterization within the circuit.

To evaluate PA performance, a coupler extracts a portion of the output signal for conversion into an intermediate frequency. This functionality is sourced from the RF blocks library.

The transmitter model encompasses additional essential elements. A baseband signal generator produces a quadrature amplitude modulation (QAM) signal as the transmitter input. This signal passes through digital interpolation filters and a digital converter, preparing it for modulation and amplification.

Addressing imperfections and distortions within the signal processing stages is a core objective. Nonlinearity, noise gain, and imbalance are considered, with digital interpolation filters and converters accurately capturing and analyzing these effects.

The PA output, amplified and modulated, connects to an external power amplifier model via RF blocks. This enables a comprehensive simulation of the signal chain, encompassing both the transmitter and power amplifier. The output feeds into the antenna for transmission.

Optimizing transmitter linearity involves an observer receiver and an adaptive Digital Pre-Distortion (DPD) algorithm. The observer receiver captures the PA output for analysis, while the adaptive DPD algorithm applies corrections to compensate for nonlinearity and related imperfections. Experimentation with various DPD strategies, filters, and power configurations refines linearity and performance.

By implementing this method, mobile signal boosting can be achieved effectively. The improved linearity of the power amplifiers reduces distortion and interference, resulting in enhanced signal quality. This, in turn, leads to better coverage, fewer dropped calls, and faster data rates, benefiting both mobile users and network operators.

In summary, the method of modeling RF power amplifiers and increasing wireless transmitter linearity with DPD using MATLAB provides a powerful approach to boost mobile signals. It enables engineers to optimize the performance of mobile communication systems by mitigating the challenges associated with power amplifier nonlinearity and improving signal quality.

Chapter 5 Testing and Analysis

5.1 Result of Method 1 - Crystal Oscillator based Mobile Signal Booster

The first method, employing a crystal oscillator, underwent thorough testing in two contrasting environments to assess its effectiveness. The initial hardware construction and evaluation occurred in Diyathalawa, where the signal strength and quality were noticeably poor. RSRP (Reference Signal Received Power) and RSRQ (Reference Signal Received Quality) was evaluated using the "field test" feature on an iPhone.

During the initial attempt, there was a distinct absence of improvement in both signal strength and quality. Subsequent iterations were performed, each yielding similar unsatisfactory results. In response, modifications were made to the antennas and circuit components in the hopes of achieving a positive outcome. However, despite multiple adaptations, no discernible difference was observed. Numerous trials ensued, involving replacements of various components and additional adjustments, yet the desired improvement remained elusive. The signal quality and strength stubbornly refused to exhibit any notable changes, whether positive or negative.

As a result, the initial test was ultimately classified as unsuccessful, prompting an examination of the circuit design, antennas, and substituted components for potential shortcomings. Despite the theoretical promise of the method, practical testing revealed its limitations in achieving the desired enhancement in signal quality and strength. Further analysis and experimentation would be necessary to refine the approach and address the underlying issues encountered during testing.

In pursuit of a solution, the second and final attempt took place in Colombo, where the signal strength and quality were notably superior. This comprehensive evaluation encompassed not only the mobile signal but also the Wi-Fi network, evaluating both download and upload speeds.

Equipped with new sets of antennas, an updated circuit diagram, and more suitable substitute components, the second attempt at implementing the signal booster commenced. The signal parameters, such as RSRP (Reference Signal Received Power) and RSRQ (Reference Signal Received Quality), were measured using the "Field Test" feature on an iPhone, while the download speed was assessed using the Speedtest by Ookla application. Despite the refined design and improved components, the outcome remained unchanged. The signal booster failed to exhibit any noticeable effect on signal strength, signal quality, or download speed.

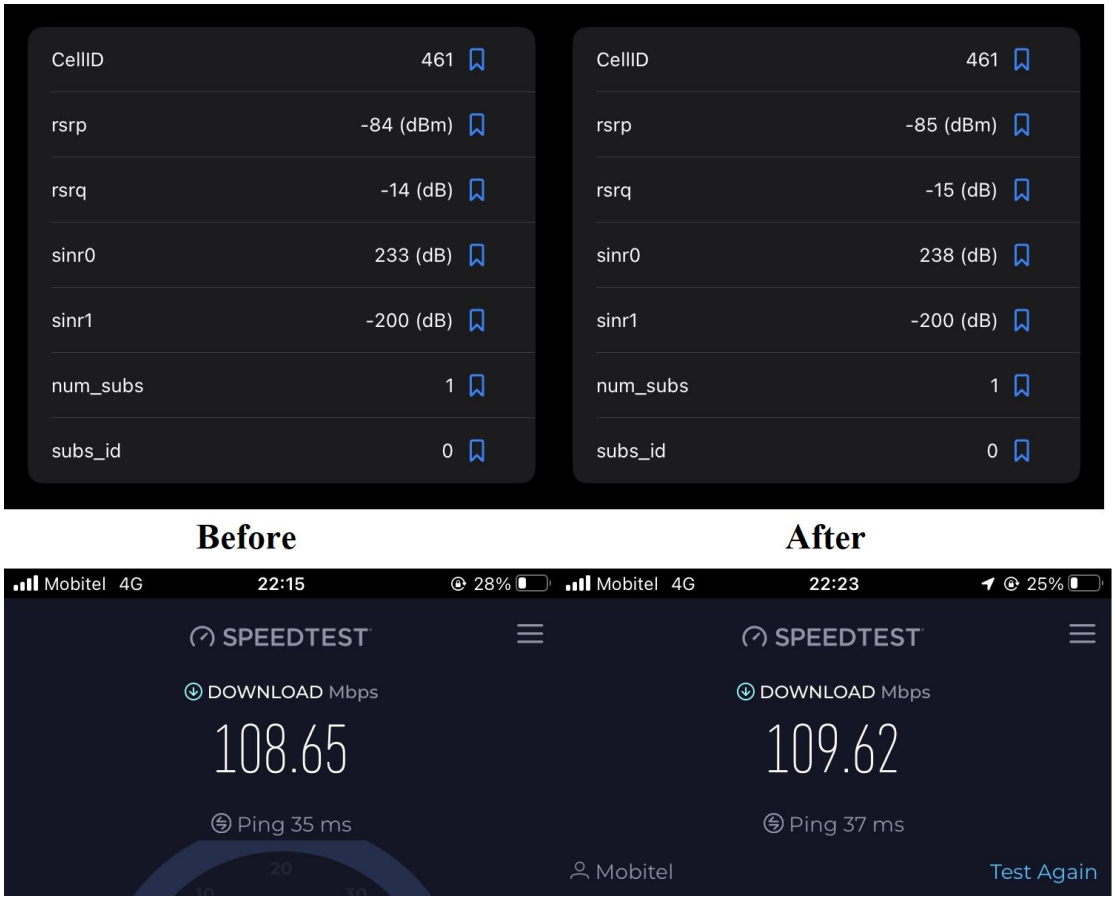


Figure 27 Test resutls before and after

Efforts were then directed towards troubleshooting to identify potential errors and rectify them. Numerous test runs were conducted, exploring various troubleshooting techniques, yet no discernible changes were observed in the mobile signal. The signal booster neither provided the desired boost in signal strength nor had any detrimental effects akin to jamming. Following this unsuccessful trial, it was concluded that this particular method was unsuccessful in achieving the intended objectives and was consequently disapproved.

5.1.1 Analyzing challenges and Implemented Solutions

Antenna - If the error is attributed to the antennas, the following method employed: a series of antenna replacements were conducted, involving the utilization of various types, sizes, shapes, and configurations with different amounts of turns. These replacements aimed to assess the impact of antenna characteristics on the signal boosting performance.

Faulty Components - In the event that the issue lies with the components, the following method was employed: a systematic component replacement approach was used, where various replaceable components were tested and substituted to identify any faulty or underperforming elements within the circuitry. If the error is traced back to the Crystal Oscillator, a plan was implemented by employing three different Crystal Oscillators. This approach aimed to explore the possibility of a faulty oscillator and ascertain whether the observed issues were specific to a particular unit.

Faulty Diagram - In the scenario where the problem was identified within the circuit diagram itself, the design implementation involved updating and modifying the existing diagram. This included revising the circuit layout, making schematic adjustments, and simplifying the diagram to enhance clarity and functionality.

A notable drawback encountered during the implementation of this method was the excessive time spent on iterative changes and modifications without exploring alternative approaches. This oversight hindered the timely progression of the project and limited the exploration of backup ideas or alternative methodologies.

Consequently, valuable time was consumed on the trial-and-error process, resulting in a delay in identifying a more effective solution or considering alternative strategies.

5.2 Result of method 2 - RF Power Amplifier and Improved Linearity with DPD to boost signal

Using the Simulink software described earlier, several simulations were conducted in MATLAB Simulink to evaluate the effectiveness of the proposed method in boosting mobile signals. The simulations focused on important parameters such as occupied bandwidth, intermodulation distortion, harmonic distortion, and adjacent channel power ratio (ACPR). By comparing scenarios with and without digital predistortion (DPD) interference, the simulations provided valuable insights into the impact of the method on mobile signal quality and interference reduction. These simulations shed light on the potential of the proposed method to enhance mobile signal transmission and improve overall performance.

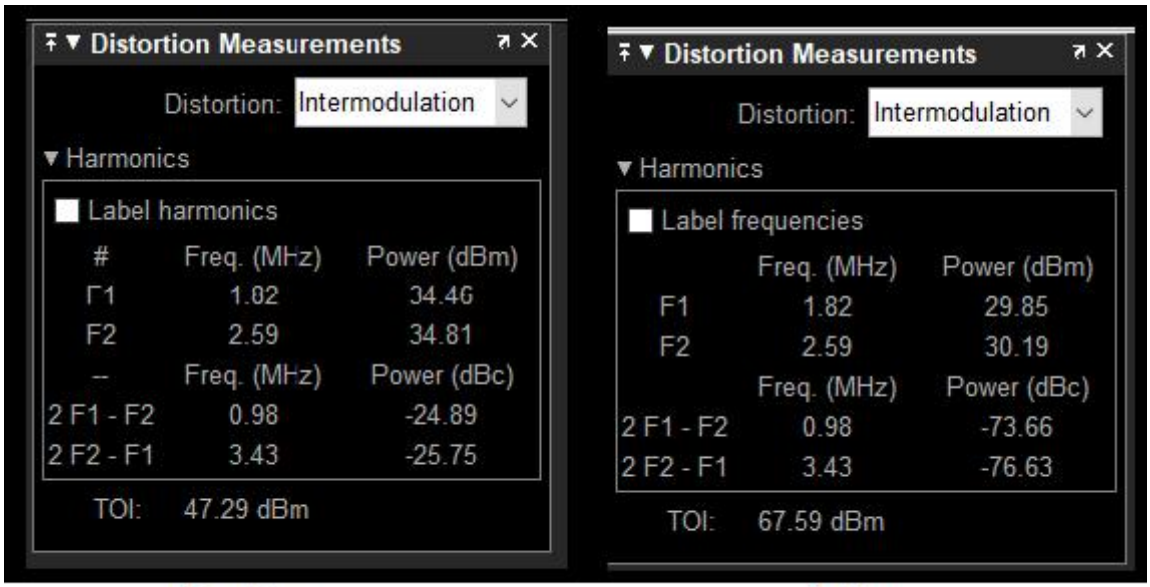
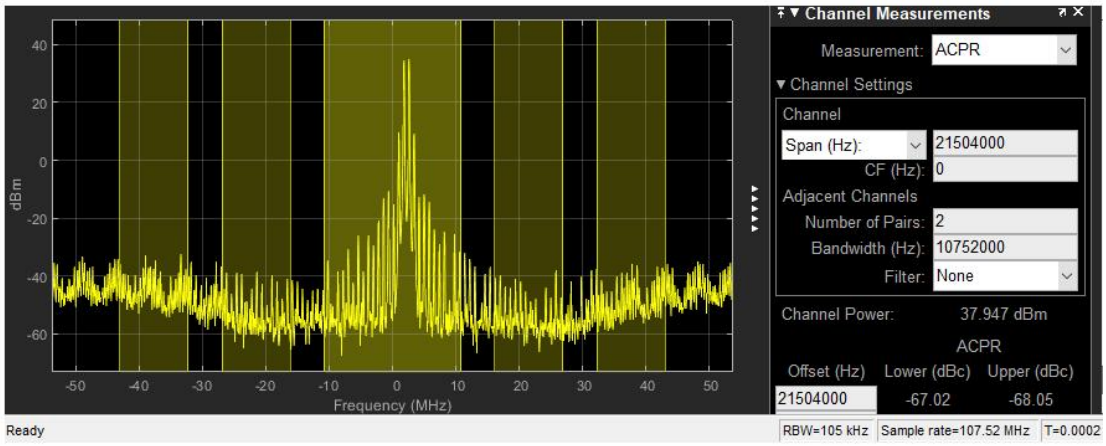
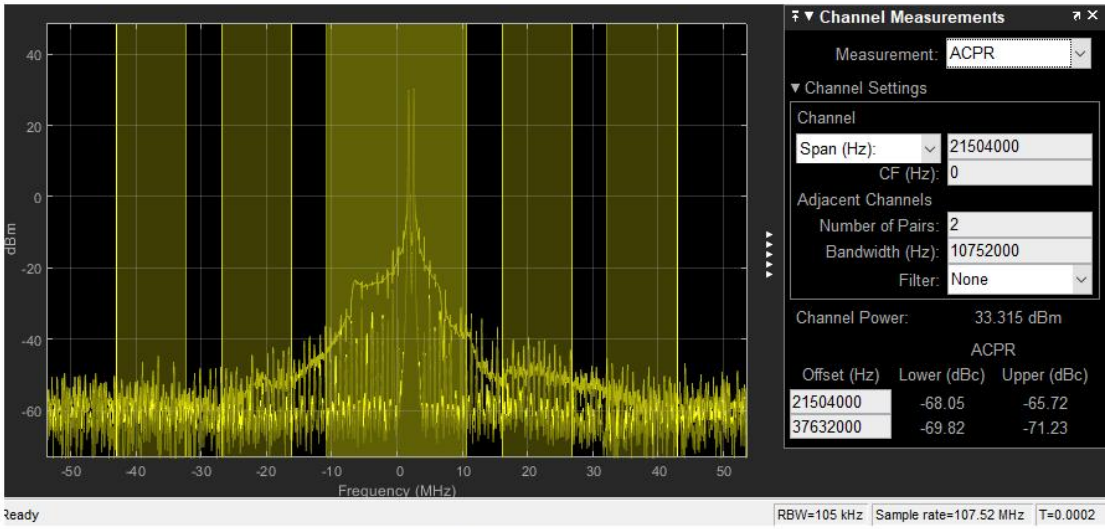


Figure 28 Toi Before and After

In the early test without digital predistortion (DPD), the third-order intercept point (TOI) in the spectrum analyzer for the transmitter was measured at 47.21 dBm. However, when DPD was applied, the TOI improved significantly to 67.59 dBm. This improvement is beneficial for mobile signal boosting because a higher TOI indicates a higher level of linearity in the transmitter. It means that the transmitter can handle stronger input signals without introducing nonlinear distortions, resulting in improved signal quality and reduced interference. Therefore, the increase in TOI with DPD demonstrates the effectiveness of the method in enhancing the linearity and overall performance of the mobile signal, making it more suitable for boosting purposes.



Before



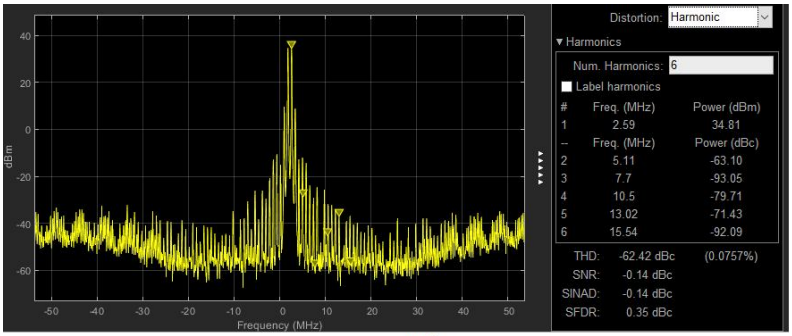
After

Figure 29 ACPR before and After

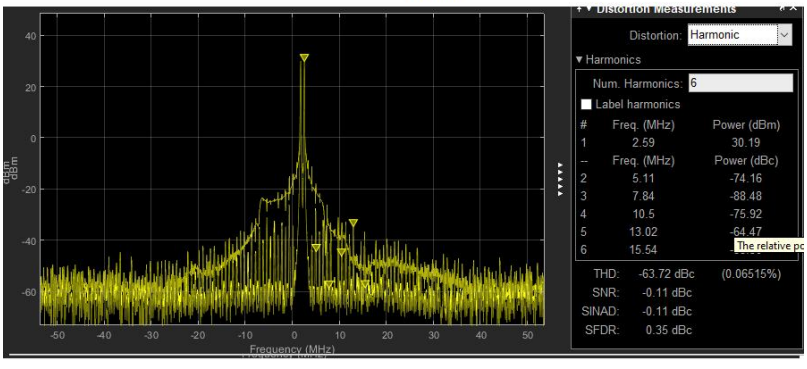
In the early test without digital predistortion (DPD), the adjacent channel power ratio (ACPR) in the spectrum analyzer for the lower and upper channels was measured at -68.05 dBc and -65.72 dBc, respectively. However, when DPD was applied, the ACPR improved to -67.02 dBc for the lower channel and -68.05 dBc for the upper channel.

This improvement in ACPR is beneficial for mobile signal boosting as it indicates a reduction in unwanted interference to neighboring channels. ACPR measures the level of power leakage from the desired channel into adjacent channels, and lower ACPR values indicate better spectral purity. By applying DPD, the transmitter can mitigate nonlinear distortions and effectively suppress unwanted spectral components, resulting in improved ACPR performance.

The reduced ACPR values with DPD demonstrate the effectiveness of the method in enhancing the signal quality and minimizing interference, ultimately leading to improved spectral efficiency and enhanced overall performance of the mobile signal. This makes it more suitable for mobile signal boosting applications where maintaining a clean and interference-free signal is crucial for reliable and efficient communication.



Before



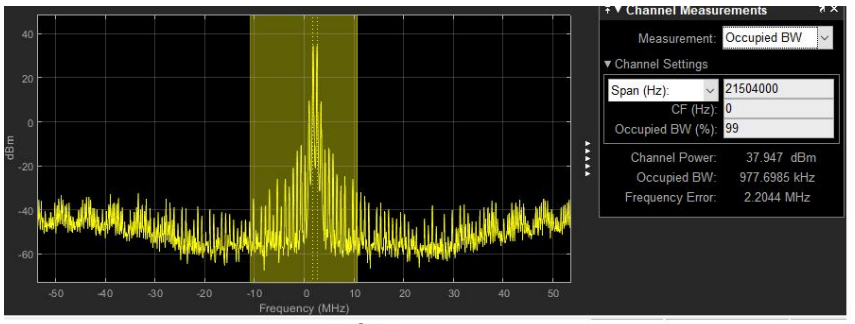
After

Figure 30 harmonic distortion before and after

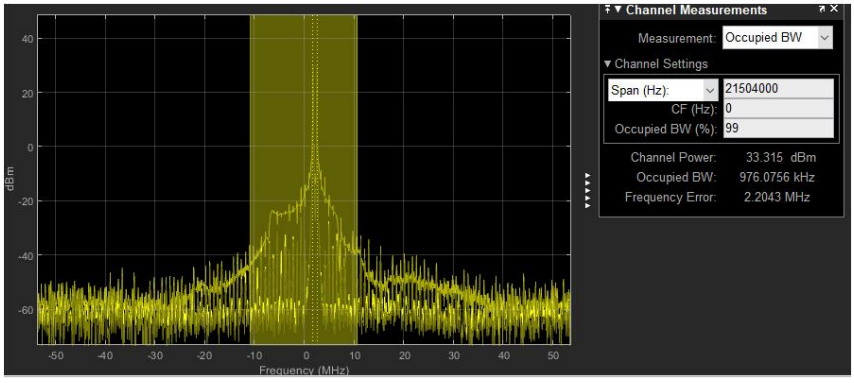
In the early test without digital predistortion (DPD), the spectrum analyzer measurements revealed a harmonic distortion level of 0.757%. However, when DPD was applied, the harmonic distortion reduced to 0.6515%.

This reduction in harmonic distortion is advantageous for mobile signal boosting as it signifies improved signal quality.The presence of undesirable frequency components that have a frequency multiple of the desired signal is referred to as harmonic distortion. High degrees of harmonic distortion can impair the integrity of the transmitted signal, resulting in interference and lowering its overall quality.

The transmitter can efficiently correct for nonlinearities and reduce the production of undesirable harmonic components by using DPD. A cleaner and more accurate representation of the original signal is implied by a decrease in harmonic distortion, which leads to an improvement in signal fidelity and clarity. This is particularly crucial for mobile signal boosting since it guarantees that the amplified signal is free from distortion and interference, improving reception, data speeds, and overall system performance.



Before



After

Figure 31 Occupied Bandwidth before and after

In the early test without digital predistortion (DPD), the occupied bandwidth measured in the spectrum analyzer TX was 977.6985 kHz. However, when DPD was applied, the occupied bandwidth reduced to 976.0756 kHz.

This reduction in occupied bandwidth is beneficial for mobile signal boosting as it signifies improved spectral efficiency. Occupied bandwidth refers to the frequency range occupied by the transmitted signal. By reducing the occupied bandwidth, more efficient utilization of the available frequency spectrum can be achieved.

A narrower occupied bandwidth allows for better frequency allocation, enabling more simultaneous communication channels within the given spectrum. This translates to increased capacity and improved overall system performance. It allows for higher data transmission rates, improved signal quality, and better utilization of available network resources.

Additionally, a narrower occupied bandwidth can help mitigate interference issues by reducing the likelihood of signal overlap with neighboring channels. This leads to enhanced coexistence with other wireless systems and improved overall network reliability.

Therefore, the reduction in occupied bandwidth achieved through DPD implementation is advantageous for mobile signal boosting, as it enhances spectral efficiency, capacity, and overall system performance, contributing to a better mobile communication experience.

Based on the obtained data and results, it can be concluded that the method of modeling RF power amplifiers and increasing wireless transmitter linearity with digital predistortion (DPD) is successful in boosting mobile signals.

Chapter 6 Conclusion and Future work

In conclusion, the primary objective of the project, which involved the hardware and software implementation of the specific targeted method, was not successfully achieved due to a multitude of challenges stemming from the limited availability and lack of clarity surrounding the resources. The project team undertook extensive efforts, encompassing the exploration of multiple designs, the creation of detailed diagrams, and the rigorous execution of numerous trials incorporating various component adaptations and replacements. However, despite these endeavors, the desired outcome could not be realized. Throughout the project's journey, considerable challenges were faced that posed significant hurdles and hindered our progress towards achieving our intended goals.

However, in light of the encountered obstacles, an alternative approach was pursued to navigate through these challenges. A different method was explored, leveraging existing technologies and techniques to address the objective of signal boosting. Through meticulous analysis and experimentation, the signal boosting task was successfully completed using this alternative method.

Unfortunately, due to hardware implementation challenges with the signal booster, the planned drone testing and calculation of device effectiveness had to be abandoned. These adjustments imposed limitations on the comprehensive assessment of the real-world impact and performance of the device. Nevertheless, it is crucial to acknowledge the substantial progress achieved despite the numerous challenges faced. A significant portion of the project was successfully accomplished, yielding invaluable experience, vital lessons, and enhanced problem-solving capabilities. The arduous journey through these obstacles undeniably contributed to professional growth and development within the realm of expertise.

6.1 Future Works

Automated Drone Integration: A promising future development for this project involves the integration of automated drones as part of the mobile signal boosting solution. By leveraging drone technology, the signal booster can be deployed in remote or difficult-to-access areas, ensuring comprehensive coverage and efficient signal propagation. Automated drones equipped with signal boosting capabilities can be programmed to autonomously navigate, identify optimal locations for signal enhancement, and dynamically adjust their position based on real-time network requirements. This integration enables rapid deployment, efficient resource allocation, and improved signal quality in challenging or underserved areas.

Integration with 5G Networks: As the telecommunications industry evolves towards 5G networks, future developments in mobile signal boosting can focus on seamless integration with the 5G infrastructure. This includes adapting the signal booster to support the higher frequency bands, massive Multiple-Input Multiple-Output (MIMO) technology, and beamforming techniques utilized in 5G networks. By optimizing the signal booster's architecture and algorithms to align with 5G specifications, it becomes possible to enhance signal quality, increase data throughput, and deliver an improved user experience in 5G-enabled environments.

Energy Efficiency Enhancement: Addressing energy efficiency is another crucial future development for this project. Advancements can be made to reduce power consumption, optimize the energy usage of the signal boosting system, and introduce energy-saving mechanisms. This can include the implementation of intelligent power management techniques, such as dynamic power allocation based on real-time network demands, the utilization of low-power components and devices, and the integration of renewable energy sources to power the signal booster. By enhancing energy efficiency, the project can contribute to sustainable and eco-friendly mobile signal boosting solutions.

These future developments hold significant potential for the advancement of mobile signal boosting technology. The integration of automated drones enables broader coverage, especially in challenging areas, while integration with 5G networks ensures

compatibility and optimization in the evolving telecommunications landscape. Additionally, the focus on energy efficiency contributes to environmentally conscious solutions, minimizing power consumption and promoting sustainable practices. Implementing these developments can result in improved connectivity, enhanced user experiences, and more efficient utilization of network resources.

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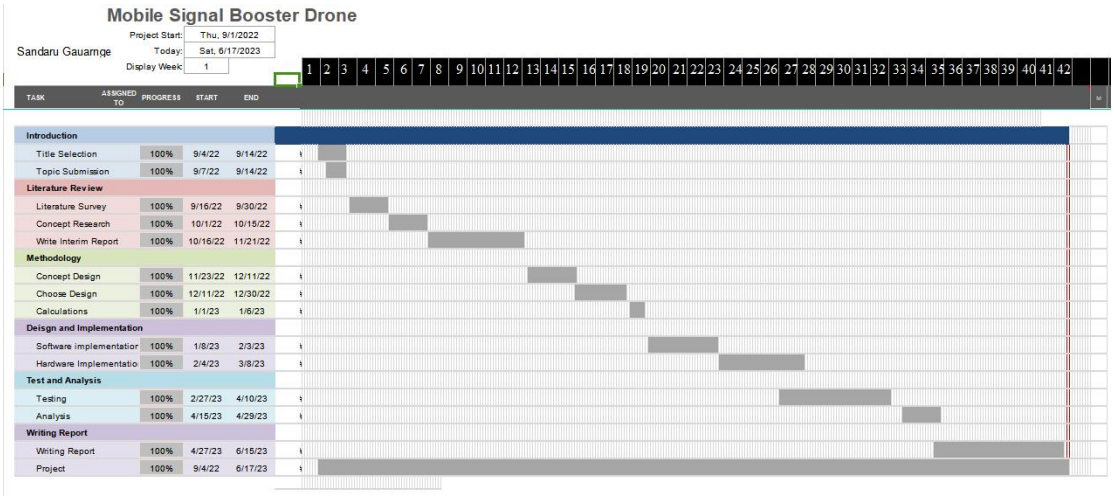


Figure 32 Gantt Chart